

GE Healthcare

Remote Monitor & Monitor-in-Room Boom Options Installation

OPERATING DOCUMENTATION



2390072-100
Rev 05

The information in this manual applies to:

Octane-Based Consoles

- H1 QX/i SDAS
- H2 LightSpeed Plus SDAS
- H3 LightSpeed (4/8) MDAS
- PET DST 16, 8, 4
- PET DST HP60

Linux-Based Consoles

- H3 w/GOC1 Console MDAS (4/8)
- H4 w/GOC1 & GOC2 Console MDAS16
- H5 w/ GRE RT, HP60, 80, 100 MDAS
- GRE Console Upgrades

IMPORTANT PRECAUTIONS

LANGUAGE

WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

警告

このサービスマニュアルには英語版しかありません。

GEMS以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。

このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。

この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意：

本维修手册仅存有英文本。
非 GEMS 公司的维修员要求非英文本的维修手册时，
客户需自行负责翻译。
未详细阅读和完全了解本手册之前，不得进行维修。
忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

DAMAGE IN TRANSPORTATION

Check for damage to property which may have occurred at the site during delivery, such as damage to floors, door frames or walls. If damage is found, notify the install specialist.

All packages should be closely examined at time of delivery. If damage is apparent, have notation "damage in shipment" written on all copies of the freight or express bill before delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14-day period.

To file a report:

- Call 1-800-548-3366 and use option 8.
- Fill out a report on <http://us44hdd21/sctq/InstallFulfill/InstalFulfillment.htm>
- Contact the local service coordinator for more information on this process.

Rev. Jan. 5, 2005

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Healthcare personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE's field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Medical Systems Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and

Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

LITHIUM BATTERY CAUTIONARY STATEMENTS

CAUTION



Risk of Explosion.

Danger of explosion if battery is incorrectly replaced.

Replace only with the same or equivalent type recommended by the manufacturer. Discard used batteries according to the manufacturer's instructions.

ATTENTION



Danger d'Explosion

Il y a danger d'explosion s'il y a remplacement incorrect de la batterie.

Remplacer uniquement avec une batterie du même type ou d'un type recommandé par le constructeur. Mettre au rebut les batteries usagées conformément aux instructions du fabricant.

OMISSIONS & ERRORS

Customers, please contact your GE Sales or Service representatives. GE personnel, please use the GEMS PQR Process to report all omissions, errors, and defects in this publication.

Revision History

Revision	Date	Reason for change
1	02/27/04	Initial Release of Manual
2	06/04/04	GOC3 (GRE) Changes 5.X Changes w/Plate
3	08/29/05	Chapter 1: Updated catalog numbers and parts lists Chapter 2: - Added information for center of gravity for Monitor-in-Room Boom kits. - Revised steps for installation of boom. - Detailed all wiring (power, ground, and video) between console, ceiling junction box, and monitor in room for both Octane and HP monitors. Chapter 3: - Added console monitor wiring overview. - Revised 4-way splitter installation. - Moved all wiring information to Chapter 2. Added Chapter 4 - Patient Touch Leakage Test
4	11/10/05	Major updates, to accommodate new mfg instructions.
5	09/05/06	Chapter 1: Updated parts lists Chapter 4: Updated procedure

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Chapter 1

Pre-Installation Information

Section 1.0

Software

There are no special software requirements. The monitors function with all software versions.

Section 2.0

Environmental Conditions

The following standard environmental condition requirements must be met. There are no changes unless otherwise noted.

2.1 Temperature and Humidity Specifications: Ambient Temperature

Maintain a temperature of 70° – 75° F (21° – 24° C) in scan room for patient comfort. When scan room is unoccupied, table and gantry temperature limitations are 64° – 79° F (18° – 26° C).

2.2 Electro-Magnetic Interference (EMI)

Color Monitor. Locate color monitors in ambient static magnetic fields of less than 5×10^{-5} tesla to guarantee color purity and display geometry.

If fields where excessive EMI is known or suspected to be present, consult GE Medical Systems for recommendations. Consider the following if you attempt to reduce EMI:

- External field strength decreases rapidly with distance from the source of the magnetic field.
- The external leakage magnetic field of a three-phase transformer is much less than that of a bank of three single phase transformers of equivalent power rating.
- Large electric motors are a source of substantial EMI.
- Steel reinforcing in building structure can be a conductor of EMI.
- High-powered radio signals are a source of EMI.
- Maintain good screening of cables and cabinets.

2.3 Seismic Data

Follow the specified seismic reinforcement requirements for your area.

Section 3.0 Catalog Numbers and Parts Lists

3.1 Catalog Numbers

CATALOG #	SYSTEM	DESCRIPTION
B7868ME	H4 - H5	Video Upgrade for Linux Console
B7530LC	H4 - H5	HS/LS Remote Color (LNX) Monitor
B7710LN	H4 - H5	LCD & SUSP (Boom in Room) Linux

Table 1-1 Catalog Numbers - Linux Systems

CATALOG #	SYSTEM	DESCRIPTION
B7710WM	Cti - H3	LCD Monitor, Boom & Suspension
B7530LC	H1 - H5	Remote Color Monitor

Table 1-2 Catalog Numbers - Octane Systems

3.2 Parts Lists

3.2.1 B7868ME - Video Upgrade for Linux

ITEM	PART NUM	QTY	FRU	DESCRIPTION
1	46-302200P10	1.0	No	1.72 X 1.10, MODEL#, SN, DATE, DESC.
2	2403438-3	1.0	Yes	5 BNC MALE TO HD 15 MALE 75 FEET
6	5161129	1.0	Yes	LCD Bracket Short Arm
7	2212717	1.0	Yes	Cable Cover for LCD Suspension 3.6 Meters Long
8	5116219	1.0	Yes	Grounding Cable For LCD Console To LCD
13	2376772	1.0	Yes	1-IN 4-OUT RGBHV DISTRIBUTION AMP
14	2403438	2.0	Yes	5 BNC MALE TO HD 15 MALE 6 FEET
16	2403438-4	1.0	Yes	HD 15 FEMALE TO HD 15 MALE 15 FEET
17	5125372-100	1.0	No	NEC 1980SXi LCD Service Manual
18	2390072-100	1.0	No	LS LINUX CRT AND LCD REMOTE MONITOR
25	2398987	1.0	No	CEILING PLATE ASSEMBLY
26	5115171	1.0	No	Handle LCD Monitor
27	2384323	10.0	Yes	75 OHM TERMINATOR
28	2384324	1.0	Yes	DVI TO HD15 CONVERTER
29	5115172	1.0	No	Ground Wire Ceiling Plate
30	46-328421P17	4.0	Yes	Screw Pan Head 4 MM 20 MM Zinc Plated, Steel
31	46-328425P1	2.0	Yes	NUT HEXAGON 4 MM
32	46-328432P2	6.0	Yes	WASHER LOCK - SPRING 4.1 MM 7.6 MM ZINC PLATED, SPRING STEEL

Table 1-3 Video Upgrade for Linux Console, Parts List (5115182 BOM, Rev 6)

ITEM	PART NUM	QTY	FRU	DESCRIPTION
33	46-328430P2	6.0	Yes	WASHER PLAIN - NORMAL 4.3 MM 9 MM ZINC PLATED, STEEL
34	46-208758P5	10.0	Yes	SELF LOCKING CABLE TIE

Table 1-3 Video Upgrade for Linux Console, Parts List (5115182 BOM, Rev 6) (Continued)

3.2.2 B7530LC Remote Color Monitor

ITEM	P/N	QTY	FRU	DESCRIPTION
1	2406043	1.0	Yes	22IN COLOR CRT MONITOR
2	2376772	1.0	Yes	1-IN 4-OUT RGBHV DISTRIBUTION AMP
3	2384322	1.0	Yes	5 BNC MALE TO HD15 MALE - 75 FT PLENUM RATED
4	2376770-2	1.0	No	5 BNC MALE TO HD15 MALE (6 FT)
5	2376771	1.0	No	5 BNC MALE TO HD15 FEMALE (6 FT)
6	2384323	10.0	Yes	75 OHM TERMINATOR
7	2384324	1.0	Yes	DVI TO HD15 CONVERTER
	2333243	1.0	Yes	21 IN Color CRT Monitor Global Standard Monitor 2002

Table 1-4 Remote Monitor Kit for Linux Console, Parts List (2377146 BOM, Rev 2)

3.2.3 B7710LN - LCD Monitor & Suspension Option for Linux

ITEM	P/N	QTY	FRU	DESCRIPTION
1	46-302200P10	1.0	No	1.72 X 1.10, MODEL#, SN, DATE, DESC.
2	2403438-3	1.0	Yes	5 BNC MALE TO HD 15 MALE 75 FEET
3	5160969-3	1.0	Yes	Complete Suspension containing ceiling column 5160969 and suspension arm 5160969-2
5	5117545	1.0	Yes	Monitor, 19 inch color LCD, silver, NEC LCD1980SXi
6	5161129	1.0	Yes	LCD Bracket Short Arm
7	2212717	1.0	Yes	Cable Cover for LCD Suspension 3.6 Meters Long
8	5116219	1.0	Yes	Grounding Cable For LCD Console To LCD
10	2213219	1.0	No	POWER CABLE FOR LCD-CONSOLE TO LCD
13	2376772	1.0	Yes	1-IN 4-OUT RGBHV DISTRIBUTION AMP
14	2403438	2.0	Yes	5 BNC MALE TO HD 15 MALE 6 FEET
16	2403438-4	1.0	Yes	HD 15 FEMALE TO HD 15 MALE 15 FEET
17	5125372-100	1.0	No	NEC 1980SXi LCD Service Manual
18	2390072-100	1.0	No	LS LINUX CRT AND LCD REMOTE MONITOR
25	2398987	1.0	No	CEILING PLATE ASSEMBLY
26	5115171	1.0	No	Handle LCD Monitor
27	2384323	10.0	Yes	75 OHM TERMINATOR
28	2384324	1.0	Yes	DVI TO HD15 CONVERTER
29	5115172	1.0	No	Ground Wire Ceiling Plate

Table 1-5 Boom-in-a-Room for Linux, Parts List (5115174-2 BOM, Rev 1)

ITEM	P/N	QTY	FRU	DESCRIPTION
30	5161926	1.0	No	Hardware collector for Boom-in-a room
	5160969	1.0	Yes	Suspension - Ceiling Mount
	5160969-2	1.0	Yes	Suspension - Boom

Table 1-5 Boom-in-a-Room for Linux, Parts List (5115174-2 BOM, Rev 1) (Continued)

3.2.4 B7710WM Exam Suite LCD Monitor & Suspension Option - Octane

ITEM	P/N	QTY	FRU	DESCRIPTION
1	46-302200P10	1.0	No	1.72 X 1.10, MODEL#, SN, DATE, DESC.
2	2216182	1.0	No	VIDEO CABLE CONSOLE TO LCD
3	2214293	1.0	Yes	CEILING COLUMN WITH FLANGE, 58CM
4	2214292	1.0	Yes	80-80 CM Extension/Spring Arm 10,3 KG (22.7 LBS)
5	5117545	1.0	Yes	Monitor, 19 inch color LCD, silver, NEC LCD1980SXi
6	5116218	1.0	Yes	LCD Bracket Short
7	2212717	1.0	Yes	Cable Cover for LCD Suspension 3.6 Meters Long
8	2213216	1.0	No	GROUNDING CABLE FOR LCD-CONSOLE TO LCD
9	2302528	1.0	Yes	VIDIO CABLE FOR LCD SUSPENSION SYST.
10	2213219	1.0	No	POWER CABLE FOR LCD-CONSOLE TO LCD
11	2213213	1.0	Yes	GROUNDING CABLE FOR LCD SUSPENSION
12	2213218	1.0	Yes	POWER CABLE FOR LCD SUSPENSION
13	2237018-2	1.0	Yes	DSC SPLITTER IN3204 4-WAY AMPLIFIER
14	2154425	1.0	Yes	SPLITTER TO CONSOLE MONITOR
15	2256485	1.0	Yes	ADAPTER, 3BNC-F TO HD15 RGB CRT
16	2154427	1.0	Yes	CABLE CARRIER TO SPLITTER
17	2223870	1.0	No	SONATA HARDWARE COLLECTOR KIT
18	2200583-100	1.0	No	CT/I (SONATA) UPGRADE MANUAL
19	2211224-100	1.0	No	LIGHTSPEED QX/I OPTIONS INSTALLATION
25	2212092	1.0	No	CEILING PLATE ASSEMBLY FOR LCD
26	5115171	1.0	No	Handle LCD Monitor
27	2316293	1.0	Yes	MONITOR CABLE ADAPTER HD15 TO BNC
28	5115172	1.0	No	Ground Wire Ceiling Plate
29	2404149	1.0	No	GROUND WIRE - CEILING PLATE

Table 1-6 Sonata Option w/LCD, Parts List (2199950-2 BOM, Rev 11)

3.3 Monitor Part Number(s)

P/N	DESCRIPTION	OPTIONS ADAPTER
5117545	LCD Color Monitor NEC	3BNC-F to HD15 FRGB CRT (2256485)

Table 1-7 Monitor Part Numbers

Section 4.0

Remote Monitor - CRT Option

Determine whether the existing cables run through conduit, wire trough or under raised floor, and if a sufficient opening exists to run the monitor video cable from the operator console to the remote reading room.

- Video Cable: 24.4 m (75 ft.) maximum length and about 5/8 in. in diameter
- Minimum 45mm (1.75 in.) hole to route cables
- Remote/slave monitor in review room (see [Figure 1-1](#))

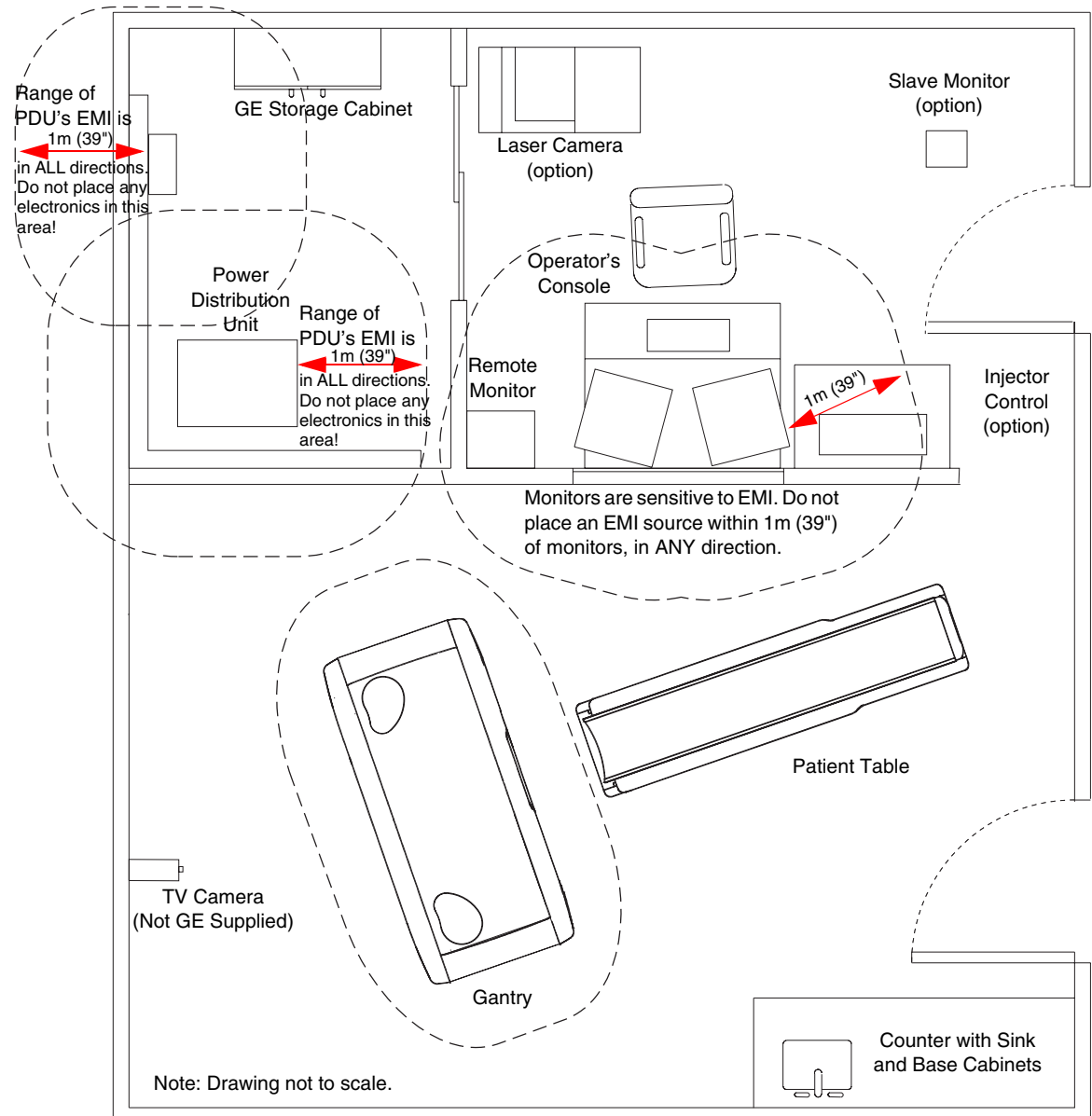


Figure 1-1 Sample room layout, showing optional remote (slave monitor) LCD/CRT

- This layout requires 3/4" conduit or existing duct from console rear wall area to the remote monitor location for video. (supplied cable)

- It also requires 3/4" to 1" conduit or existing duct from console rear wall area to remote monitor location for monitor power. (supplied cable)
- The #8 ground is required. Route the #8 ground from the raceway to the monitor. The ground runs with the power wires. (supplied cable)

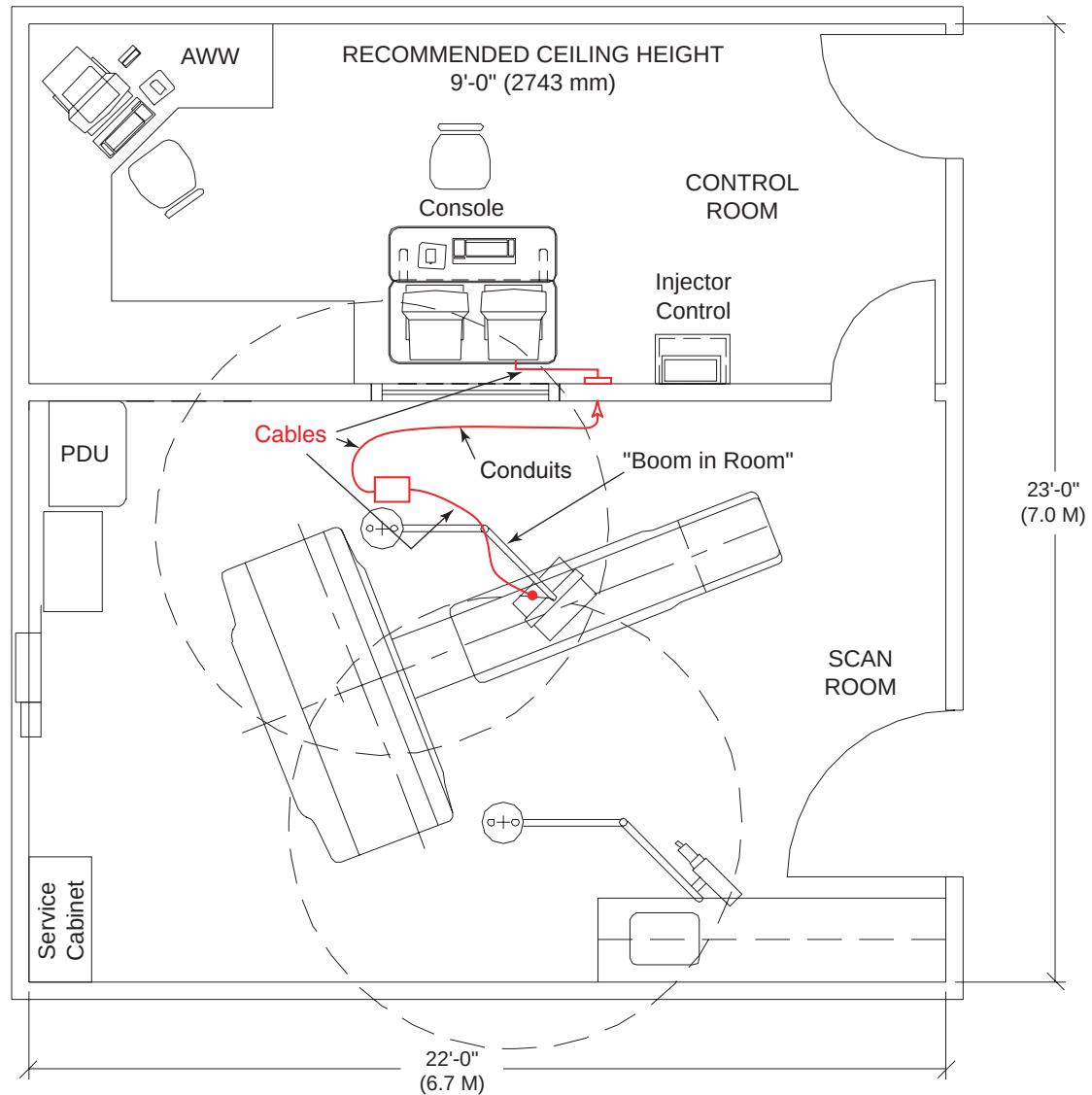


Figure 1-2 Standard Room Layout, Showing Area of Boom Travel

- This layout requires 3/4" conduit or existing duct from the back of the console area to the ceiling or wall mount plate for boom in room.
- The layout also requires 1" conduit for power from the console back area to the wall plate power plug.
- The #8 ground runs from the raceway to the console to the ceiling plate. The ground runs with the power wires.

Section 5.0

Exam Suite Monitor with LCD Option

5.1 Site Preparation

Determine whether the existing cables run through conduit, wire trough or under raised floor, and, if a sufficient opening exists to run the monitor video cable from the operator console to the gantry base.

- Video Cable: 24.4 m (75 ft.) long and about 5/8 in. in diameter
- Minimum 45mm (1.75 in.) hole to route cables between Console and Gantry
- LCD Monitor

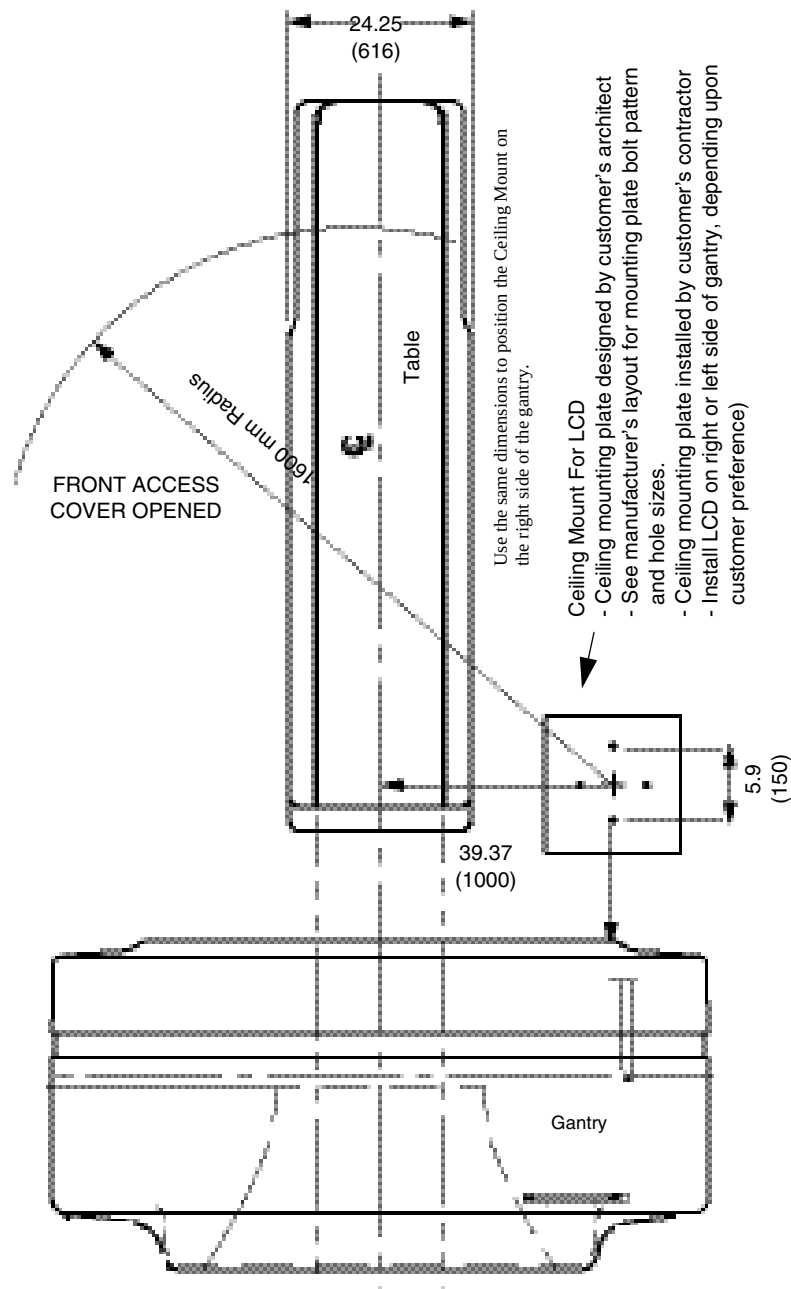


Figure 1-3 LCD Monitor

5.2 Component Dimensions

MODEL/Part Num.	5117545
WIDTH	406mm (16")
DEPTH	215mm (8.5")
HEIGHT/LENGTH	429mm (16.9")
WEIGHT	8kg (17.6 lbs)

Table 1-8 LCD Monitor Dimensions



NOTE: DIMENSIONS ARE IN INCHES (MILLIMETERS).

NOTE: NOT TO SCALE + ALL DIMENSIONS APPROXIMATE

(Follow installation drawings to position the Ceiling Mount.)

NOTE: MAKE SURE TO ALLOW FOR ADEQUATE SERVICE CLEARANCE

Figure 1-4 Table and Gantry (Top View) Exam Suite Monitor with Ceiling Mounted LCD

5.3 Power Requirements

The monitor receives its power from the Operator Console.

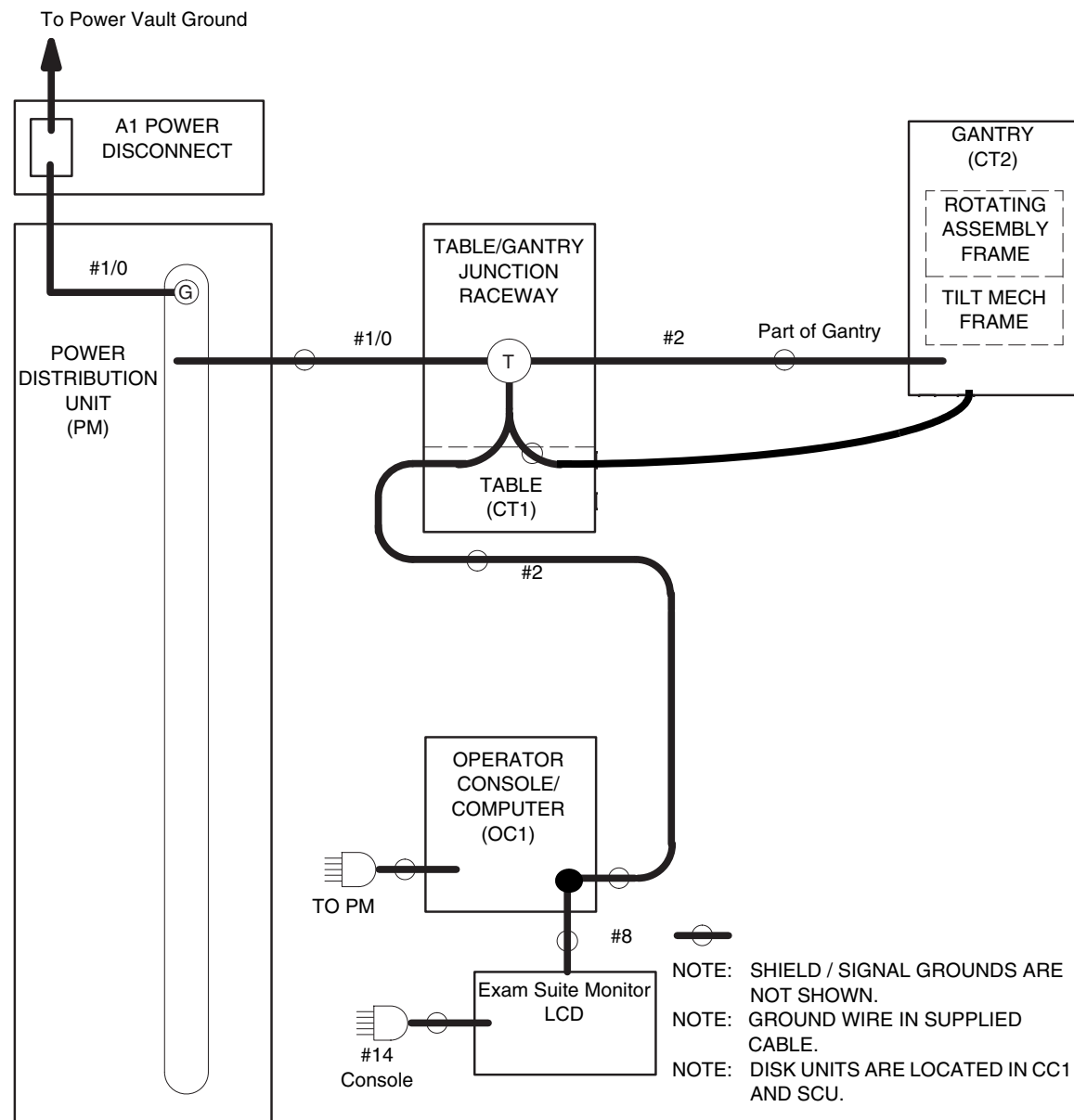


Figure 1-5 Monitor Power and Ground Map

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Chapter 2

Boom Installation

Section 1.0

Introduction

Follow the manufacturer's shipped boom and arm installation directions to attach the pedestal assembly to the customer installed ceiling mounting plate.

- Assemble and install the pedestal to the ceiling mounting plate (customer installation responsibility) using the supplied hardware.
- With the pedestal column installed, follow the manufacturer's shipped directions to install the extension spring arm and LCD monitor mounting post.
- Attach the supplied LCD monitor post using the manufacturer's shipped directions.
- Attach the LCD monitor to the post using the directions found in this manual.
- Install the cables and cable cover between the monitor and the ceiling junction box. Connect this wiring to the ceiling junction box.
- Route the LCD power, ground, and video cables to the rear of the operator console. Connect this wiring to the ceiling junction box.
- Connect the power cable to the console's outlet box.
- Connect the ground cable to the console grounding lug.

Section 2.0

Required Tools

Be sure these tools are available before installing the boom.

Step Ladder, 6' - 8'	Side Cutters
3/8" Ratchet Set (SAE/Metric)	Wire Stripper
Allen Wrench Set	Dale 600 meter (or equivalent)
Screwdrivers, Phillips and Standard (+ and -)	Fluke 87 meter (or equivalent)

Table 2-1 Required Tools

Section 3.0

Installation Tips

Use the manufacturer's supplied instructions and the following tips, to install the **ceiling pedestal**:

- 1.) When using the supplied pedestal mounting template, orient the template so that the writing on the page (i.e., bottom of page) is toward the gantry. Notice that the safety chain attachment tab will be on the right side of the print.
- 2.) The customer's architect is responsible for designing a structure and plate with sufficient strength to hold the boom and monitor assembly.
- 3.) Five (5) mounting holes are required to install this new pedestal to the ceiling mounting plate. Four (4) M12 with flat washers, lock washers, and hex nut are used. One M8 is used to anchor the safety chain bracket assembly. See [Figure 2-1](#) and [Figure 2-2](#).

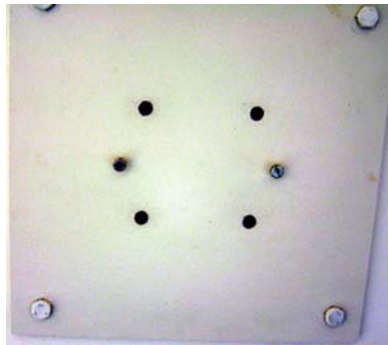


Figure 2-1 New Ceiling Plate

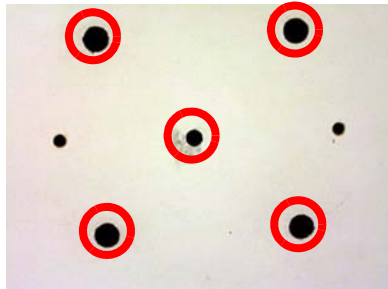


Figure 2-2 Mounting Plate (5 mounting holes are circled)

Ceiling pedestal (4) 12mm x 60mm and the safety chain (1) 8mm x 50mm.

- 4.) The pedestal cover is attached with screws to allow the cover to be lowered for installation. If a power cord is present, it must be cut before installing the pedestal. See [Figure 2-3](#).



Figure 2-3 Ceiling Pedestal

Determine the proper orientation for the arm assembly before mounting the pedestal to the ceiling plate. See [Figure 2-4](#).



Figure 2-4 Boom Orientation

- 5.) When attaching the safety chain ([Figure 2-5](#)), be sure the M10 bracket hardware is properly torqued. The torque for the M8 is 19.2N-m (14.2 ft-lbs).

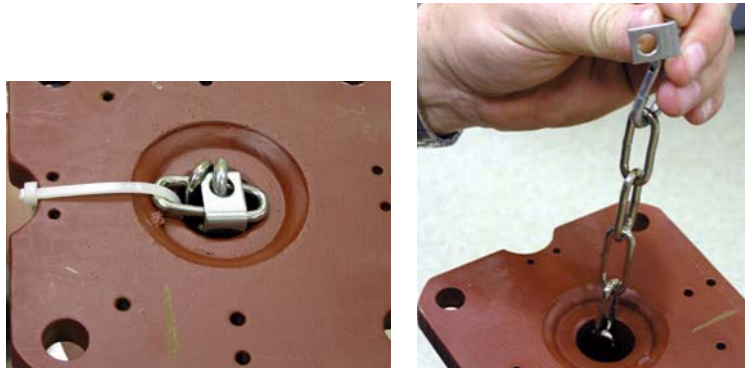


Figure 2-5 Safety Chain

- 6.) The pedestal is installed using four (4) M12 bolts. Install a flat washer over the bolt, then insert the bolt through the pedestal and ceiling plate. Add another flat washer, a lock washer and hex nut. Torque the M12 bolts to 66.4 N-m (49 Ft-lbs). See [Figure 2-6](#).



Figure 2-6 Pedestal Installed

- 7.) Install the ceiling pedestal cover, but leave the ceiling open — a ground wire will be added later. Install the bottom pedestal cover by snapping it together. See [Figure 2-7](#)



Figure 2-7 Pedestal Cover

Use the manufacturer's supplied instructions and the following tips, to install the **arm assembly**:

- 1.) After removing the covers from both the pedestal mount and the arm assembly, install the arm using the manufacturer's supplied procedures.



Figure 2-8 Hardware for attaching arm



Figure 2-9 Attaching Arm to Pedestal



Figure 2-10 E-ring Tool

- 2.) The monitor mounting bracket should fit snugly into the arm assembly. Be sure to follow the supplied manufactures instructions when completing this assembly.



Figure 2-11 Install LCD Monitor Post

There is a new spring tab that is mounted above the keeper to keep the locking tab in place. Be sure this assembly is correctly assembled.



Figure 2-12 Locking Tab and Screw

Use the manufacturer's supplied instructions and the following tips, to complete the **electrical connections**:

- 1.) The electrical box mounted in the ceiling should be next to the pedestal. The box should have two conduits into the box that run between the console and the ceiling.



Figure 2-13 Ceiling Box

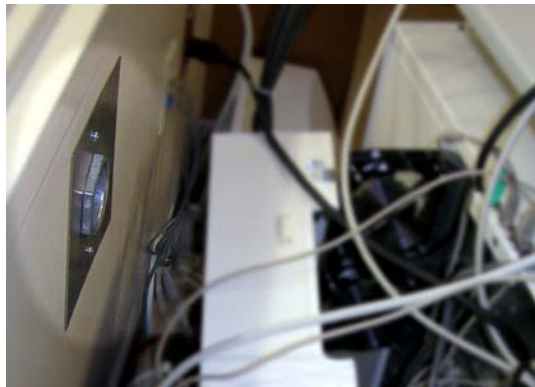


Figure 2-14 Wall Box (behind console)

The finished ceiling should look similar to that shown below.



Figure 2-15 Finished Ceiling Plate

- 2.) Some installers may choose to install the video splitter in the console on the right side. Route the cables from the wall box into the console.

Use the manufacturer's supplied instructions and the following tip, to complete the **arm balancing**:

- When balancing the arm, be sure to have all cables routed on the assembly, and the cable dressed in the supplied cable cover. Leave enough of a loop to allow full vertical and horizontal motion when done.
- Adjustment locations are shown in the figures below.

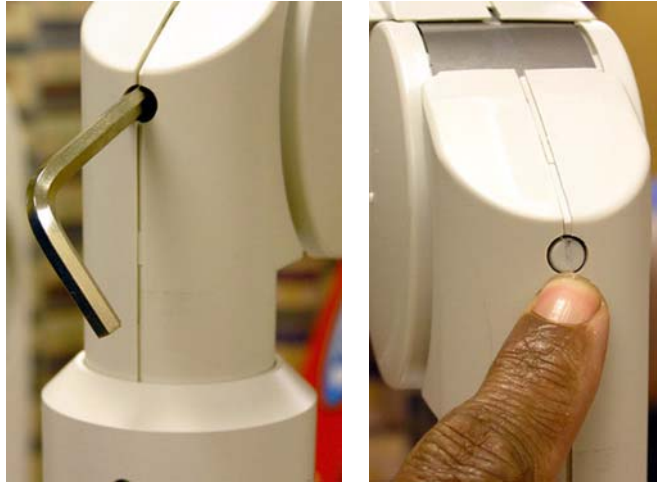


Figure 2-16 Vertical Adjustment

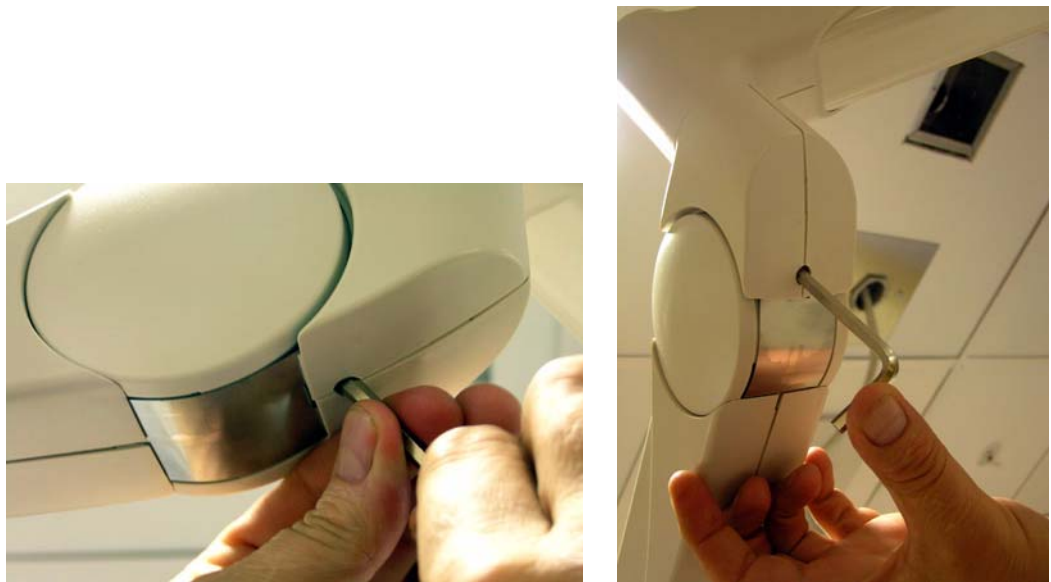


Figure 2-17 Balance Adjustment

The **finished assembly** should look similar to that shown below.



Figure 2-18 Completed Assembly



Figure 2-19 Completed Assembly - Pedestal close-up

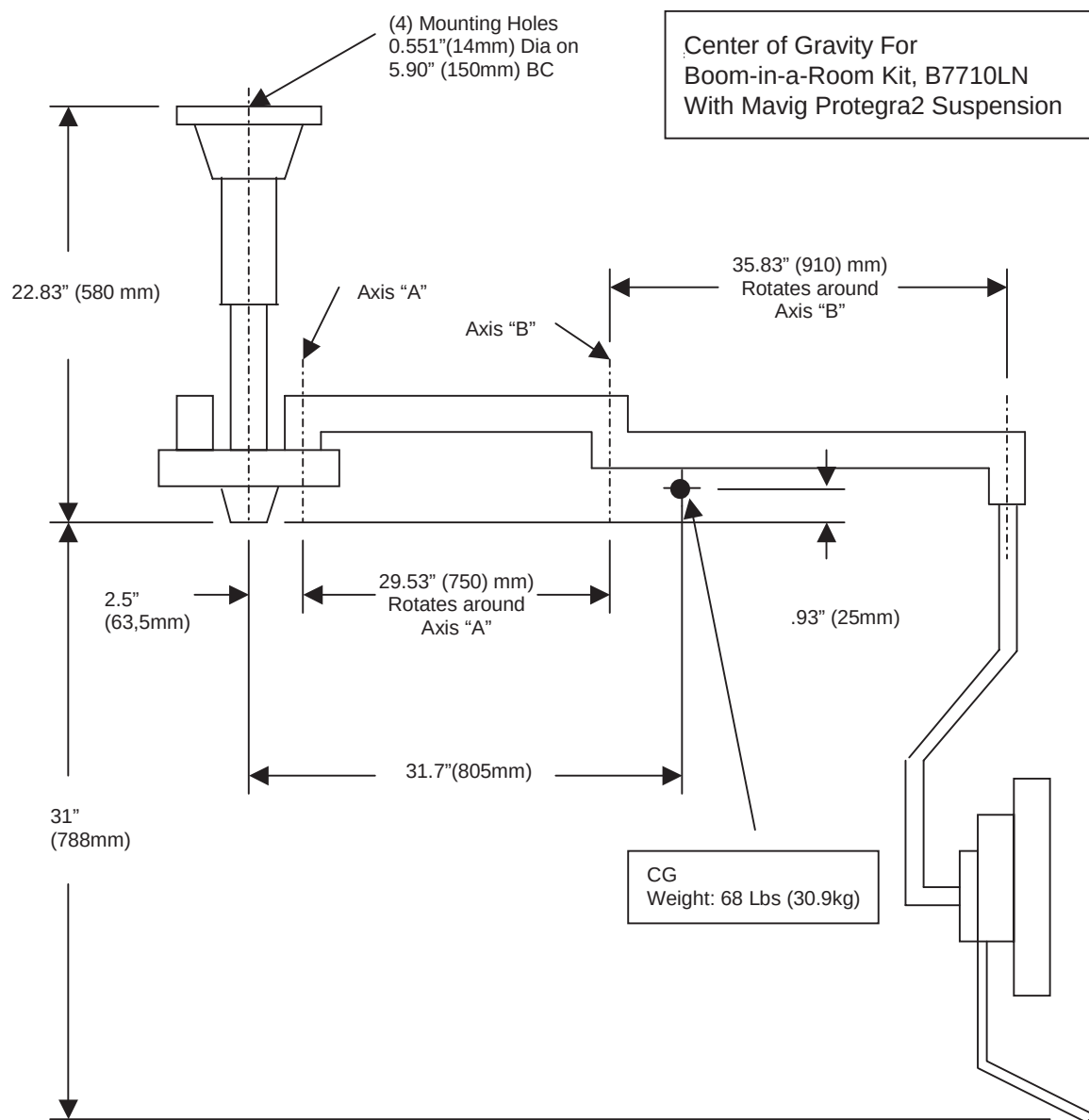


Figure 2-20 Center of Gravity for Boom-in-a-Room Kits

Section 4.0 Before You Begin

- Review local electrical codes regarding cable routing and electrical box cover plates.
- Order the appropriate kit for your system. (Review applicable catalog numbers in [Chapter 1.](#))
- Identify the remote monitor location.

DANGER



ELECTROCUTION HAZARD

HIGH VOLTAGES PRESENT.

USE PROPER LOCK-OUT/TAG-OUT PROCEDURES BEFORE WORKING ON EQUIPMENT.

WARNING



DANGER OF PERSONAL INJURY

INJURY OR DEATH CAN OCCUR FROM FALLING FROM LADDER.

FOLLOW ALL SAFETY PRECAUTIONS WHEN WORKING ON A LADDER.

WARNING



CRUSH HAZARD

HEAVY COMPONENTS CAN FALL, CAUSING PERSONAL INJURY OR DAMAGE TO EQUIPMENT.

USE TWO PEOPLE TO INSTALL THE BOOM.

Section 5.0 Procedure

5.1 Power Down System

DANGER



ELECTROCUTION HAZARD

WORKING ON ENERGIZED EQUIPMENT CAN RESULT IN ELECTROCUTION OR DEATH.

ALWAYS TAG AND LOCK OUT POWER TO THE SYSTEM AT THE “MAIN” DISCONNECT. DO NOT APPLY POWER TO THE SYSTEM UNTIL ALL WORK IS COMPLETE AND ALL COVERS REPLACED.

Before performing any of the installation procedures described in this section, do the following:

- 1.) Power down the system.
- 2.) Perform Lock-Out/Tag-Out.
- 3.) Remove the console front cover.
- 4.) Locate the power panel.
- 5.) Remove the back cover if required.

5.2 Install Ceiling Column

Refer to the manufacturer's installation instructions to complete this task. (The titles and number of steps/sections listed are for Revision ACB03O04 — these may vary for other revisions.)

- 1.) Review Technical Specifications section.
- 2.) Complete "Drilling Template for Ceiling Installation and Safety Anchor" (five sections).
Locate the full page drilling template and place on the mounting plate.
- 3.) Complete "Installing the Extension/Spring Arm Combination to the Ceiling Column" (six steps).
- 4.) Perform "Installing the Flat Panel Monitor (FPM) and FPM Adapter..." until the plastic caps have been placed into the drilled holes on the side of the safety collar (first 12 steps only).
When the plastic caps have been inserted, return to this Direction to install the LCD monitor (see section 5.3, below).

5.3 Attach Monitor

WARNING

CRUSH HAZARD



HEAVY COMPONENTS CAN FALL, CAUSING PERSONAL INJURY OR DAMAGE TO EQUIPMENT.

USE TWO PEOPLE TO ATTACH THE MONITOR.

Following the instructions below, assemble the LCD bracket, the monitor handle, the LCD monitor, and the ground wire as shown in [Figure 2-21](#).

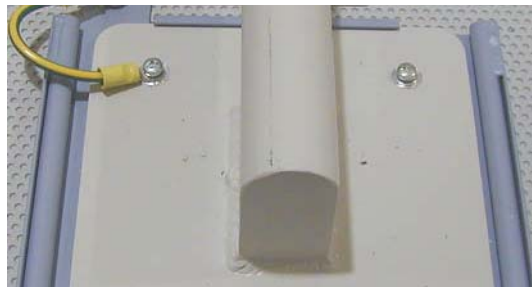


Figure 2-21 Completed LCD Monitor, Handle, Bracket, Ground

- 1.) Remove the monitor stand from the monitor.
- 2.) Assemble the bracket, handle, and monitor using the four mounting screws as shown in [Figure 2-21](#). For each of the four mounting holes, apply a thin bead of red Loctite 242 to one of the M4x20mm mounting screws and then install the screw, a flat washer and a lock washer as shown in [Figure 2-22](#).



Figure 2-22 M4x20 Mounting Screw (shown installed at right)

5.4 Route and Connect Monitor Cables

- 1.) Route the cables along the boom arm through cable cover as shown in [Figure 2-23](#). Leave enough slack at each joint to permit the full range of motion without stretching individual cables.

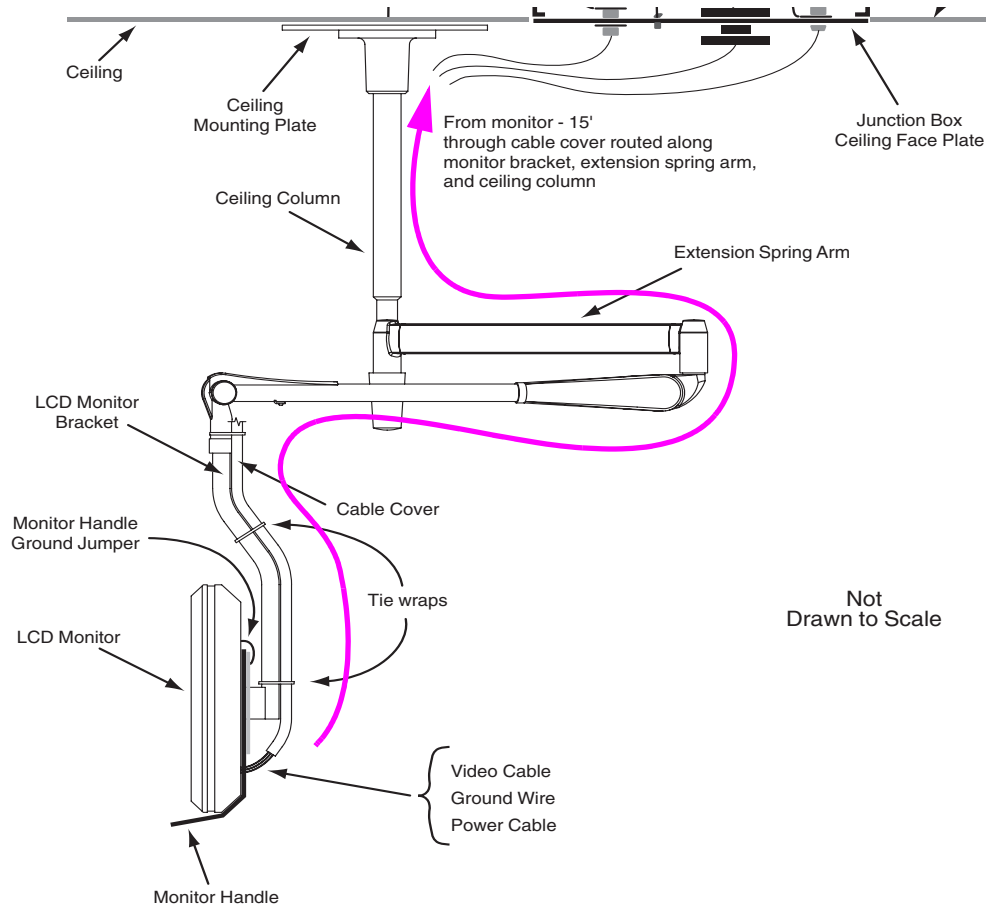


Figure 2-23 LCD Monitor Assembly with Cables and Cover

- 2.) Connect the ground cable and the ground wire jumper to the ground stud on the monitor handle bracket as shown in [Figure 2-24](#).

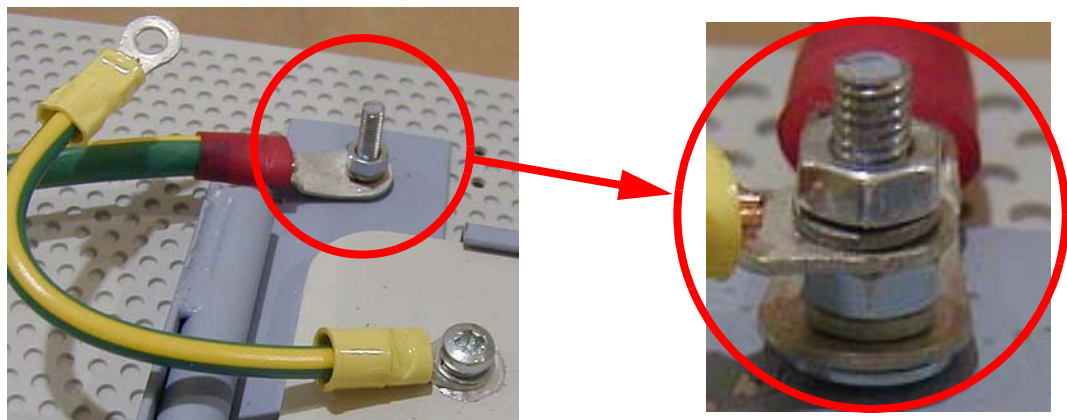


Figure 2-24 Monitor Grounding Wire

- 3.) Attach the video cable and the power cable to the rear of the LCD monitor as shown in [Figure 2-25](#).

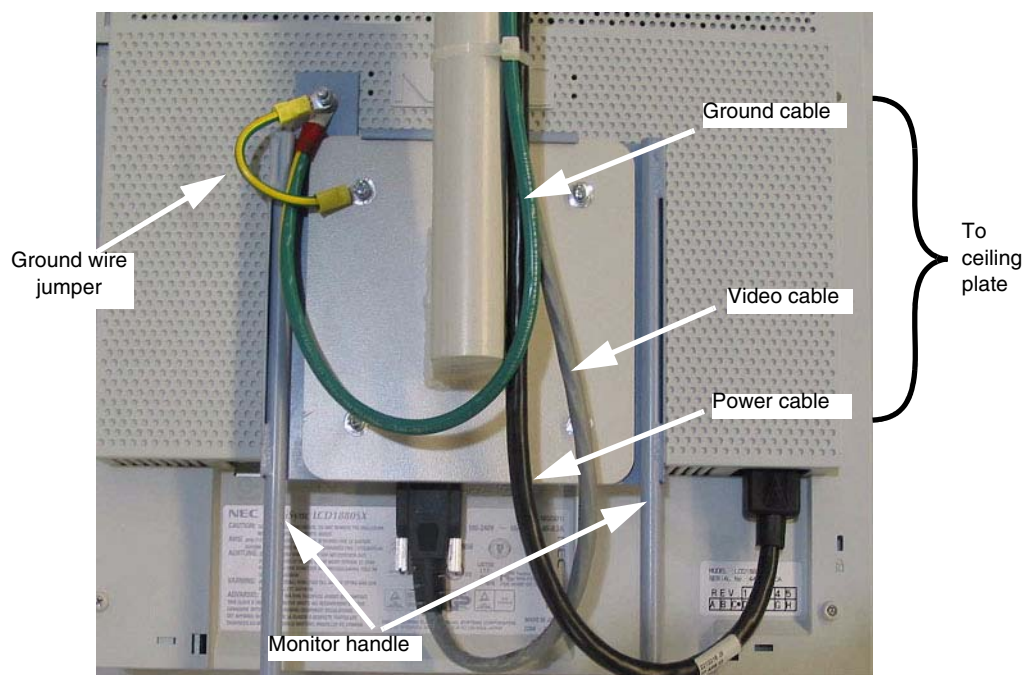


Figure 2-25 Monitor Cabling

5.5 Connect Cables at Ceiling Junction Box

DANGER ELECTROCUTION HAZARD



HIGH VOLTAGES CAN CAUSE INJURY OR DEATH.

USE PROPER LOCKOUT/TAGOUT PROCEDURES BEFORE WORKING ON EQUIPMENT.

5.5.1 Install Ceiling Face Plate

Note: The cables require installation of conduits and face plates. Consult local electrical codes. Refer to the installation site print for additional conduit size and length details.

- 1.) Select the appropriate ceiling plate to install on the remote monitor ceiling electrical box. One plate for Octane-based system has BNC connectors, while the HP version has one HD15 connector.

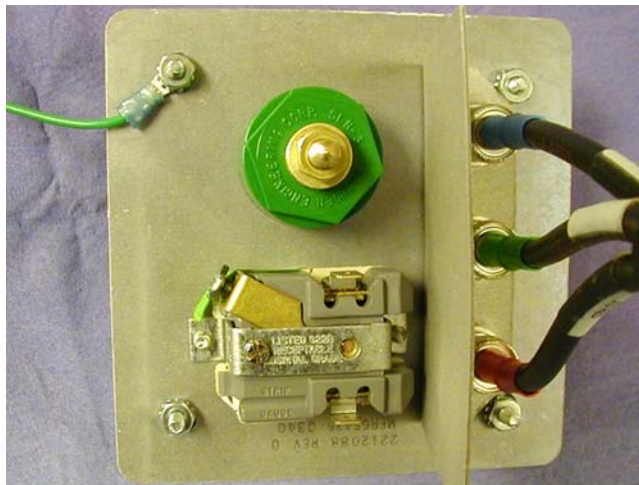


Figure 2-26 Ceiling Face Plate - Octane System (Rear View)

- 2.) **For HP Systems Only:** Prepare the ceiling plate as follows.
 - a.) The ceiling face plate kit is shown in [Figure 2-27](#). Detach the plastic bag with small parts and remove the connector tabs.



Figure 2-27 HP Ceiling Face Plate Kit

- b.) Place the connector tabs onto the face plate in the location for the video cabling. Firmly slide them into place so that the slot in each tab fits over the plate and their holes line up with the holes in the face plate. (See [Figure 2-29](#).)

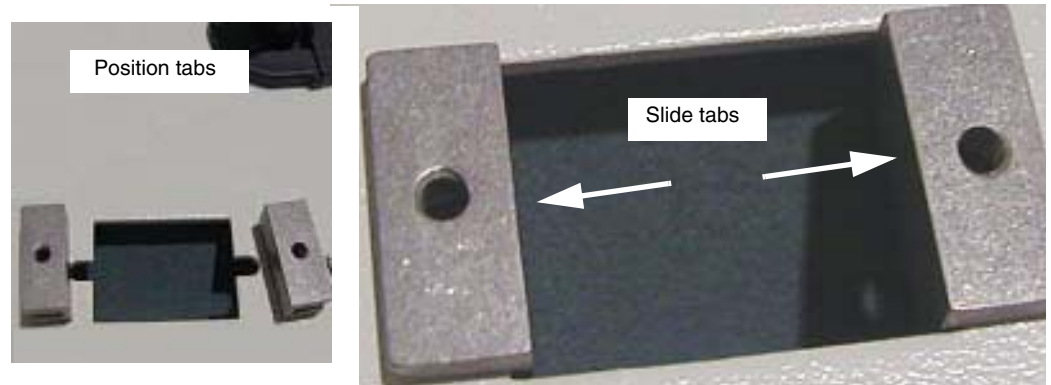


Figure 2-28 Installing Video Connector Tabs

- 3.) Connect the video, power, and ground cables from the room monitor to the ceiling face plate, as shown in [Figure 2-29](#).

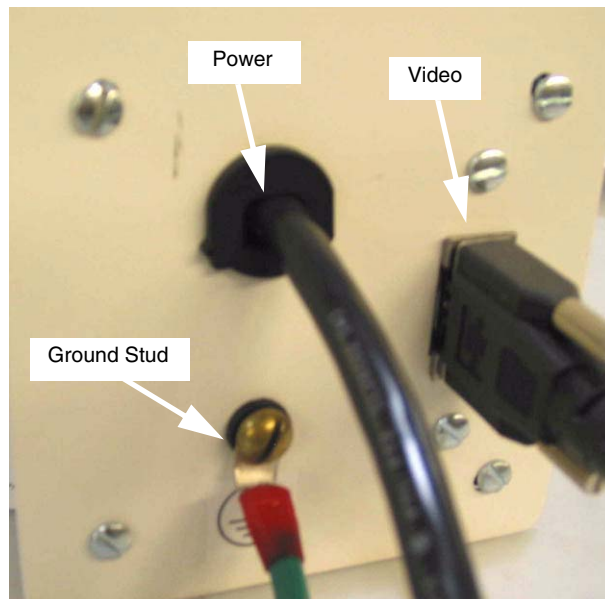


Figure 2-29 Ceiling Face Plate - HP System - Front View (Octane not shown)

5.5.2 Connect Power Cable

- 4.) Route the power cable between the operator console and the junction box for the room monitor as follows. (Maximum length for the power cord is 70' and for video cables, it is 75'.)

Note: There are no optional cables available.

- a.) Strip the power cord as shown in [Figure 2-30](#).

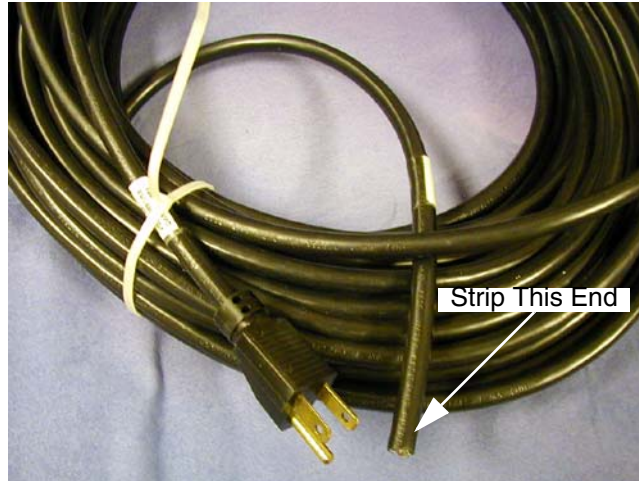


Figure 2-30 Power Cord for Octane-Based System (HP cord not shown)

- b.) Route the cable through the face plate behind the console first before pulling wires to the ceiling face plate junction box.
- c.) Route the power and ground wires to the remote ceiling junction box.
- 5.) Install and connect the power cable in junction box. (See [Figure 2-31](#) and [Figure 2-32](#).)

Note: For HP, connect the power cable to corresponding wiring using the three wire nuts supplied. Use the appropriate table below, or equivalent wiring, as a guideline for power connections.

FROM CONSOLE		TO LCD MONITOR
Black	Hot	Brass terminal
White	Neutral	Silver terminal
Green	Ground	Green

Table 2-2 Power Connections for Octane

FROM CONSOLE		TO LCD MONITOR
Black	Hot	Black or Red wire
White	Neutral	Blue or White wire
Green	Ground	Green wire

Table 2-3 Power Connections for HP

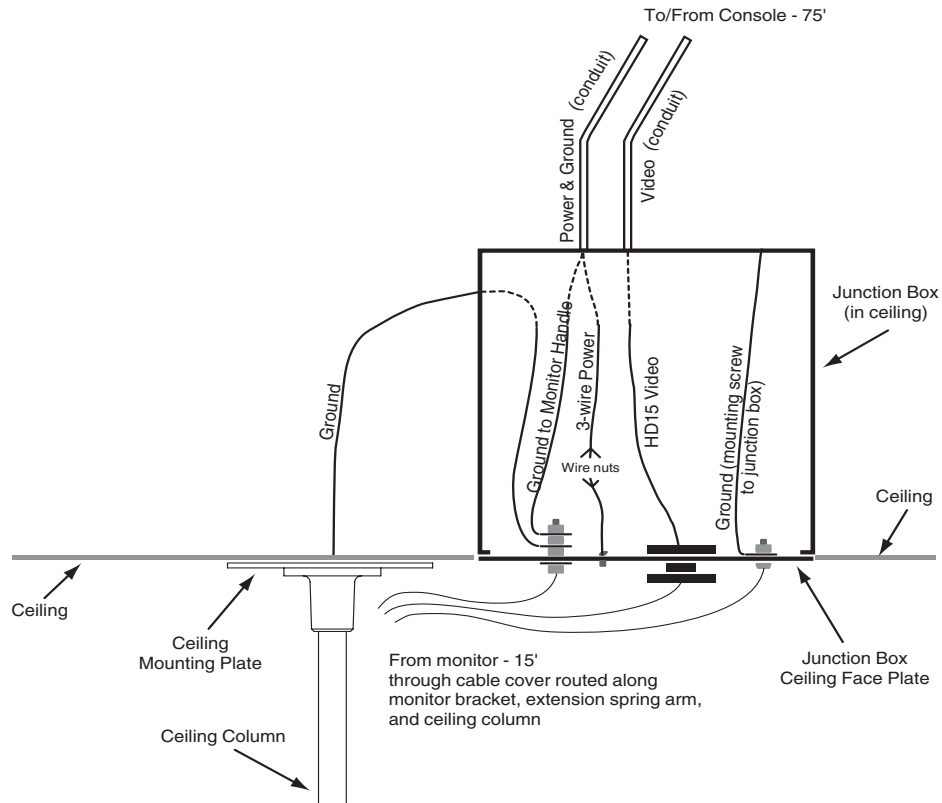


Figure 2-31 Monitor Cables at Ceiling Face Plate

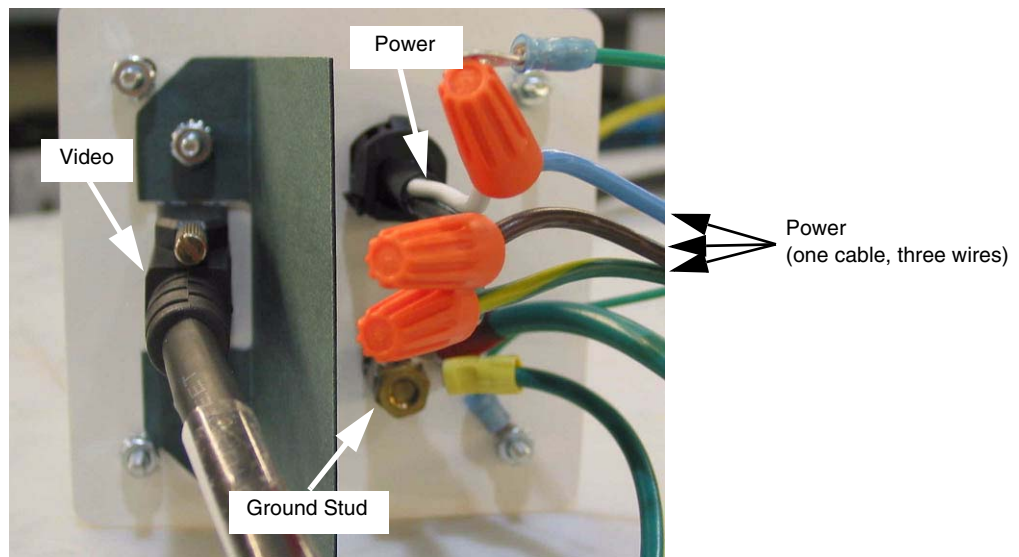


Figure 2-32 Ceiling Face Plate - HP System - Back View (in junction box)

5.5.3 Connect Ground Wires

- 1.) Connect #8 ground wire to the ground stud.



Figure 2-33 Supplied Ground Wire With Ring Terminals

GROUND STUD	
1 #8	From console
1 #10	From ceiling column
1 #14	Face plate to junction box

Table 2-4 Ground Stud Connections

- 2.) Connect the ceiling column ground and the console ground to the ground stud on the ceiling face plate.

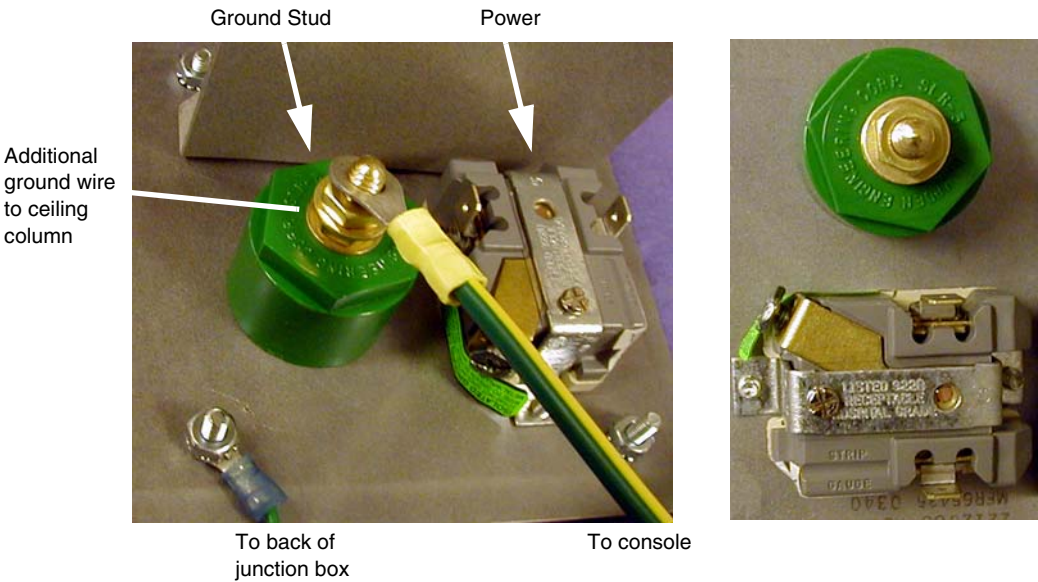


Figure 2-34 Ceiling Face Plate Ground Connection - Octane System

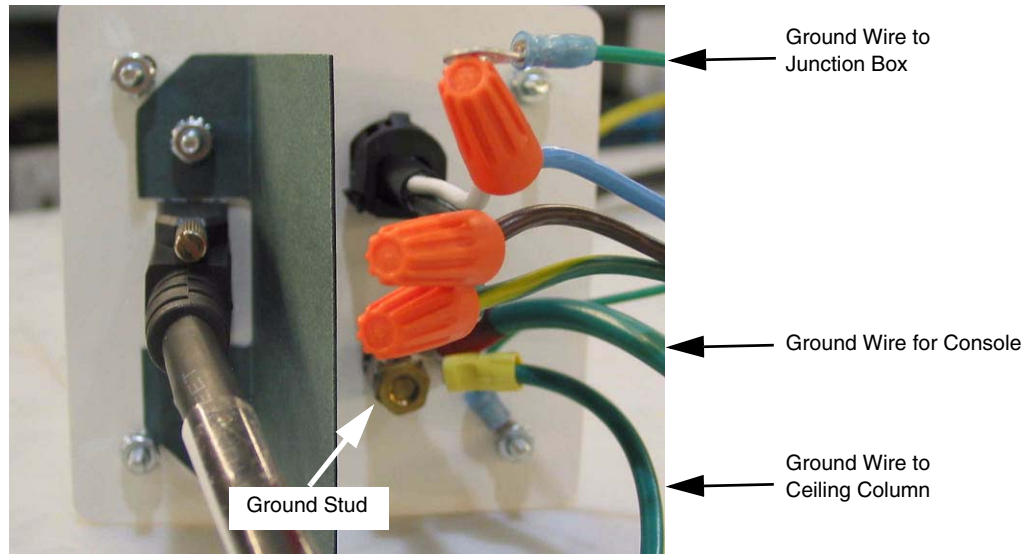


Figure 2-35 Ceiling Face Plate - HP System - Back View (in junction box)

5.5.4 Connect Video Cabling

NOTICE The video converter wires are very fragile and break easily. A pull wire is required for pulling this cable in conduit.

- 1.) Tape a pull wire to the BNC end of the video cable. (See [Figure 2-36.](#))

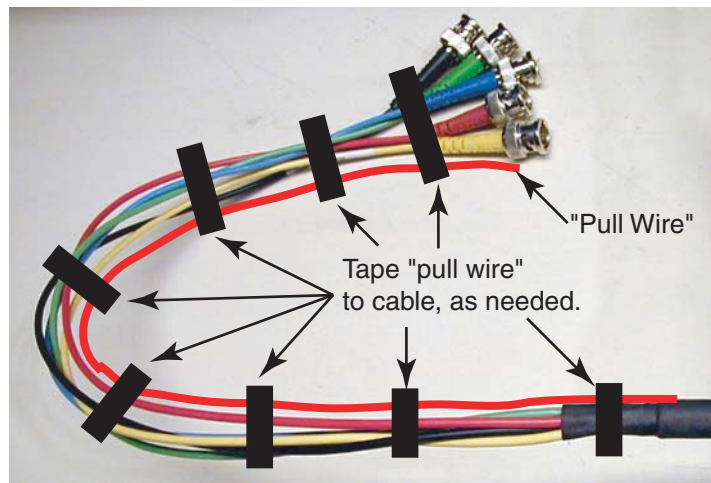


Figure 2-36 HP Remote Video Cable With "Pull Wire"



Figure 2-37 Octane Remote Video Cable (2302638)

- 2.) Using the pull wire, very carefully pull the video cable through separate conduit from power.
 - For HP, pull the video cable *from the junction box* at the remote monitor location to the splitter at the console.
 - For Octane, pull the video cable *from the splitter* at the console to the remote junction box.
- 3.) Connect the video cable at both ends.

5.5.5 Complete Wiring Installation

- 1.) Neatly arrange all the wire connection inside the ceiling electrical box.
- 2.) Attach the ceiling face plate ground wire to the electrical box.
- 3.) Attach the ceiling face plate to the ceiling junction box with the four supplied screws.
- 4.) Attach the #6 ground wire to the boom mounting plate. This wire is attached to the ceiling face plate.

Section 6.0 Balance Extension Spring Arm

Refer to the manufacturer's installation instructions to complete this task. (The titles and number of steps listed are for Revision ACB03O04 — these may vary for other revisions.)

- 1.) Complete "Adjusting the Height of the Spring Arm." (Three steps.)
- 2.) Complete "Spring Arm Weight Adjustment." (Three steps.)
- 3.) When finished with the spring arm weight adjustment procedure, return to this Direction, to continue with the installation (see [Chapter 3](#)).

Chapter 3

Wiring at Console

Section 1.0

Console Monitor Wiring Overview

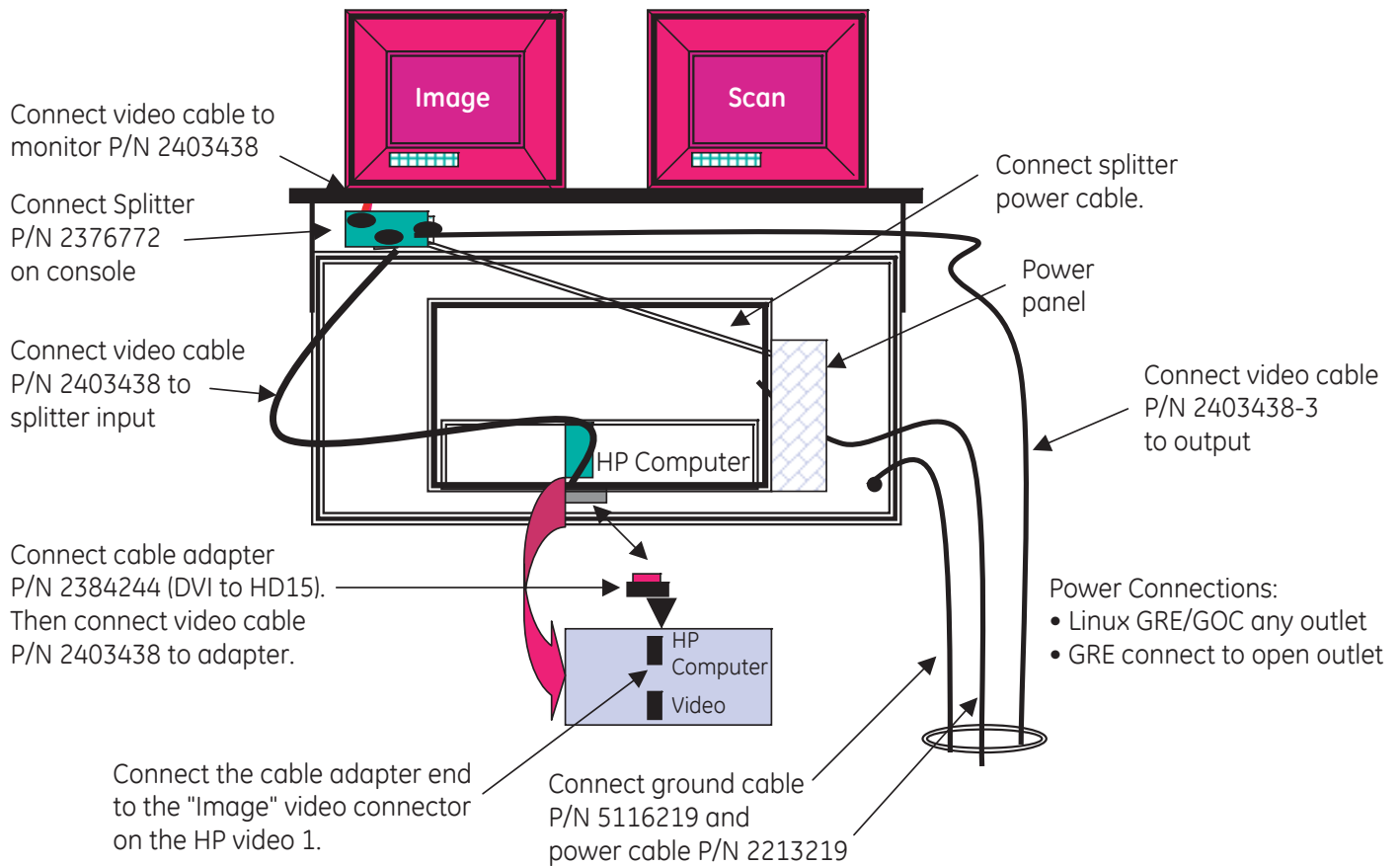


Figure 3-1 View from Behind Console

Section 2.0 Wiring Installation

2.1 Power Down System

DANGER ELECTROCUTION HAZARD



HIGH VOLTAGES CAN CAUSE INJURY OR DEATH.

USE PROPER LOCKOUT/TAGOUT PROCEDURES AT THE “MAIN” DISCONNECT BEFORE WORKING ON EQUIPMENT.

CAUTION



Do not apply power to the system until all work has been completed and all covers are in their proper place.

Before performing any of the installation procedures described in this section, do the following:

- 1.) After the customer has saved all information: Power down system.
- 2.) Perform LOTO at A1 breaker.
- 3.) Remove the front console cover, using a screwdriver for the two quarter-turn screws located on the console bottom. Screws may differ for your console type.
- 4.) Locate the console power panel.
- 5.) Remove the console back cover if required to install the splitter.

2.2 Install 4-Way Splitter

Follow the steps below to install the 4-way splitter.

Figure 3-2 shows splitter connections when completed.

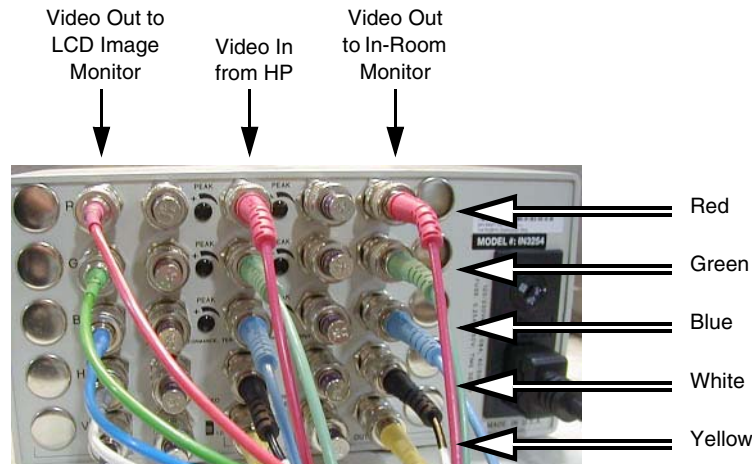
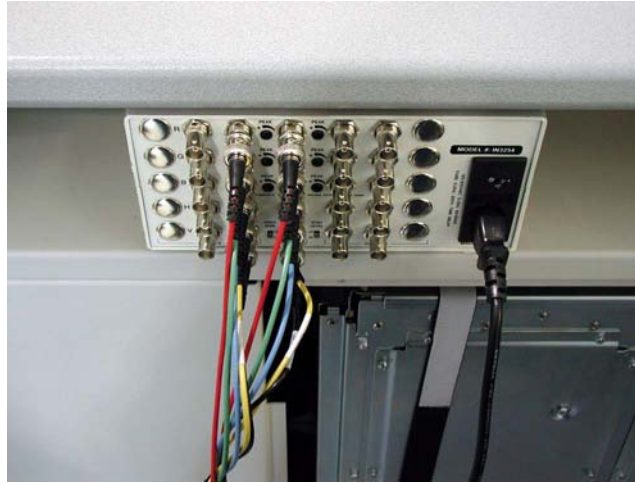


Figure 3-2 4-Way Splitter Connections, Completed

Note: The video cable connections shown in Figure 3-2 apply to all 4-way splitters, even if the splitter itself appears different from that shown.

- 1.) Determine where to locate the 4-port video splitter. There are several options to choose from:
 - Mount the splitter between the console box and the monitor table top with Velcro tape.



- Mount the splitter on the monitor table top behind the SCSI tower on GOC consoles or behind the monitors on all other consoles with Velcro tape.



- Mount the splitter inside the console on the right side if there is no DASM option.



- 2.) With the monitor and computer switched off, remove the video cable from the back of the image monitor. See [Figure 3-3](#). (This is the cable that comes from the HP or Octane.) Remove this cable from the Octane or HP computer video connector. Discard this cable. If the monitor has BNC connectors, remove them.

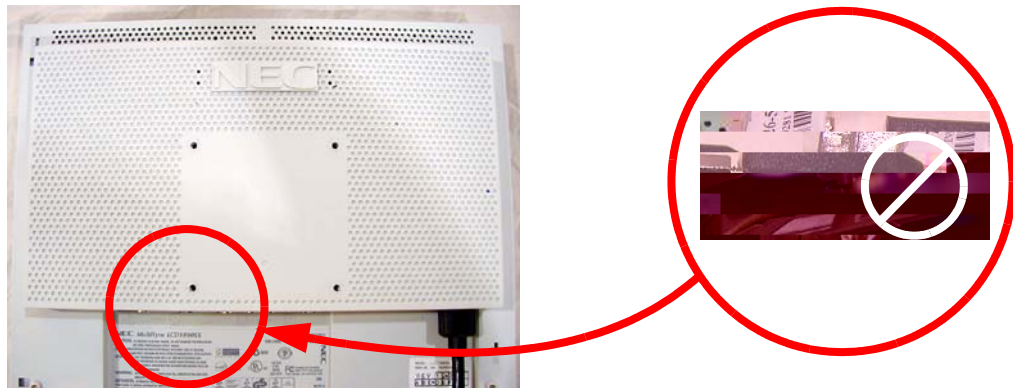


Figure 3-3 Image Monitor (with Video Connector at right)

- 3.) **For the HP computer**, add the DV1-to-HD15 adapter to cable 2403438 and connect to the HP video 1 port. See [Figure 3-3](#)

Figure 3-4 DV1-to-HD15 Adapter

- 4.) Connect the appropriate supplied video cable to the Octane or HP computer ([Figure 3-5](#) and [Figure 3-6](#), respectively) video connector and then route the cable to the 4-way video splitter. Each kit has the required cable for the connection between the computer and splitter input. Secure the cable with supplied tie wraps.

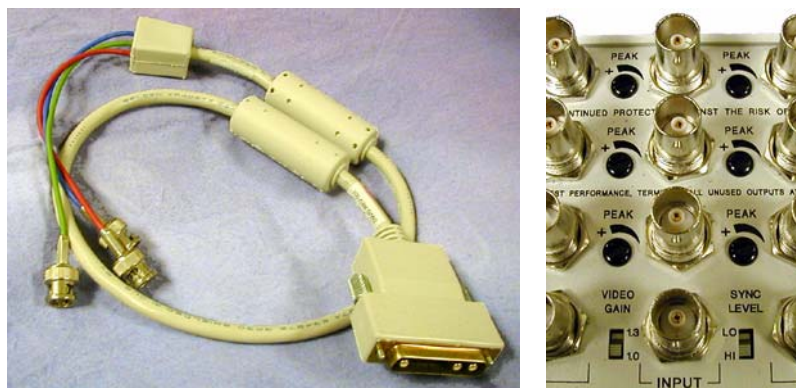


Figure 3-5 Octane Kit - Cable (left) and Splitter (right)

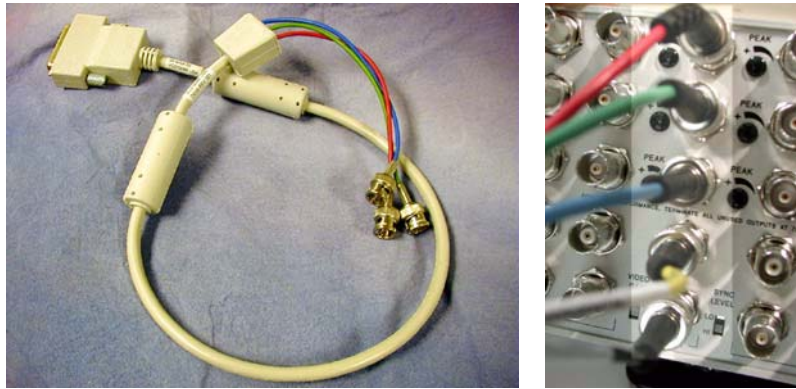


Figure 3-6 HP Kit - Cable 2403438 & Adapter (left) and Splitter (right)

NOTICE



- 5.) Route the BNC end of the video cable to the center row of BNC connectors on the video splitter. Use caution when connecting BNC cables. Observe color codes and name on cables.
Potential for Equipment Damage. Touching the video signal cable connector pins may cause them to bend. When connecting the video signal cable, check the alignment of the HD15 connector. Do not force the connector in the wrong way.
- 6.) From the kit, select the second short video cable (the one that has a BNC on one end and HD15 on the other end) and connect this cable between the video splitter and the image monitor.
- 7.) Add the video cables to the video splitter output side.
For Octane installations, the video cables have only three BNC connectors (Red Green Blue.) HP systems have five BNC connectors.
 - a.) Ensure the cables are attached to the video inputs (from above).
 - b.) Attach the cables from the desktop image monitor to the video outputs on the left.
 - c.) Attach the cables from the remote monitor to the video outputs on the right.
 - d.) Add BNC terminators to unused outputs.

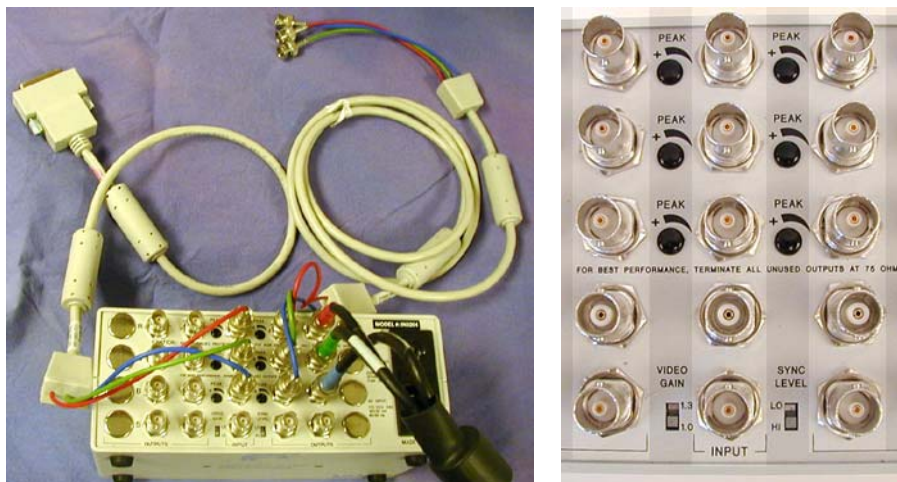


Figure 3-7 Splitter connections - Octane (left) & HP (right)

- 8.) Set the Video Gain Level to 1.0 and the Sync Level to HI.
- 9.) Mount the video splitter in your selected location and secure with 2" Velcro tape as needed.

- 10.) Select any open outlet and attach the AC power cable from the splitter to the power panel. (See [Figure 3-8](#).) Connect the remote power cord to any open outlet. If you have removed the rear console cover, reinstall that cover now.



Figure 3-8 Console power panel

- 11.) Connect one end of the ground wire with the ring terminal to the ground lug on the back of the console. See figure below for GOC and GC console.



Section 3.0

HP Video Upgrades (from Octane)

Note: Be sure to use kit B7868ME. Do not substitute individual parts for kit.

Remove old components and replace with new components:

- Replace 3-BNC splitter with new 5-BNC splitter
- Install new 6' monitor cable to video splitter
- Install new 6' video cable between the video splitter and the HP (attach the adapter to the HP)
- Install new 70' 5-BNC video cable between the splitter and the monitor's HD15 connector (NOTE: There is NO video connection at the ceiling plate)

Section 4.0

Auxiliary Monitors

Use instructions with the wall bracket to mount the wall bracket assembly.



Figure 3-9 Wall Mounted Remote Monitor

For desktop, find a suitable location to install the remote monitor.

- 1.) Neatly arrange the video and power cables along a designated route between the rear of the Operator Console and your remote location.
- 2.) If your installation requires the use of a wall face plate, follow the steps in the above procedure, now.
- 3.) See [Figure 3-10](#), below, to connect the video and power cords to your monitor. **DO NOT** connect the remote ground to the monitor.



Figure 3-10 Back of LCD Monitor

- 4.) Refer to [Section 2.2](#) for instructions on connecting the video splitter (Octane or HP system).

Section 5.0 Finalization

5.1 Restore System Power

- 1.) Install all covers.
- 2.) Check all electrical connections.
- 3.) Remove LOTO.
- 4.) Turn ON power to the system.

WARNING



POSSIBLE ACCIDENTAL SHOCK TO PATIENTS.

INCORRECTLY-POWERED AND GROUNDED WIRING CAN CAUSE ELECTROCUTION, SERIOUS INJURY OR DEATH.

CAREFULLY CHECK ALL GROUND CONNECTIONS, THEN PERFORM THE LEAKAGE/GROUND CURRENT TEST PROCEDURE IN THE FOLLOWING SECTION.

5.2 Adjust Monitor Contrast/Brightness

Note: Use the LCD monitor installation manual to set up the monitor first.

This section provides the basic procedure necessary to set up and adjust a new or replacement monitor. The monitor's manual (located in the monitor box) should be referenced during this procedure.

- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.
- 2.) Click on the IMAGE QUALITY icon.
- 3.) Select INSTALL SMPTE IMAGE and wait approximately 3-4 minutes for SMPTE image to install. When complete, the following message will display:
SMPTE and QX images have been successfully copied
- 4.) Press ENTER to exit the Service Desktop.
- 5.) Click the IMAGEWORKS icon.
- 6.) Display the SMPTE pattern. Use the browser to select Exam 1000, which contains the SMPTE pattern, and enlarge the image to full screen display.
- 7.) Select VIEWER.
- 8.) Select 1:1 FORMAT.
- 9.) Increase the monitor's contrast to maximum.

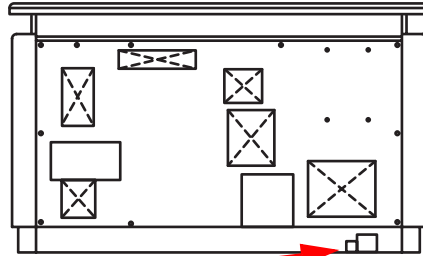
Note: Adjust monitor contrast until the operator sees the anatomical structure (window raster).

- 10.) Increase the Brightness to maximum.
- 11.) Decrease the Brightness until the raster just fades into—and matches—the monitor screen background. At this point, the 5% and 95% patches should be just visible.
 - If additional tweaking is required to attempt to match the monitor image to the filmed image, use only the brightness control.
 - If the CRT image exhibits any tearing or smearing of the alphanumeric characters, then reduce the contrast setting slightly until the tearing/smearing is just eliminated. The optimum setting for contrast is the highest setting that does not cause tearing/smearing of the alphanumeric characters.
 - Always finish up by displaying and filming images of anatomy (typical heads and bodies), and asking the technologist to compare the CRT image to the film image.

- 12.) To set up the text (left) monitor:
 - Swap the right and left monitor's video cables, to view the SMPTE pattern on the left monitor.
 - Repeat steps 2 - 11.

5.3 Attach Console Modification Plate

- 1.) Remove the modification plate from the upgrade kit.
- 2.) Refer to [Figure 3-11](#). Attach the new plate to the left of the rating plate(s). Take care not to cover any part of the existing plate(s).



Attach MOD Plate to the left of the existing plate

Figure 3-11 Attach Modification Plate to Operator Console

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Chapter 4

Patient Touch Leakage Test

Section 1.0 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	10 mins	20 mins	10 mins

Section 2.0 Overview

This procedure must be completed after all options have been installed. It covers three safety and leakage current checks:

- Patient Touch current test (completed after installation)
- System Ground resistance measurement (completed during installation)
- Ground current typical (completed after installation - optional)

Section 3.0 Preliminary Requirements

3.1 Tools and Test Equipment

Item	Qty
Standard service tool kit	1
Dale 600 Meter (from Tool Pool)*	1
Dale extended length leads*	1

*This procedure has been validated only with the Dale 600. GE Healthcare cannot guarantee the accuracy of this procedure, if any other meter is used.

3.2 Safety



WARNING

POTENTIAL FOR SHOCK

GROUND WIRES WILL HAVE GROUND CURRENT PRESENT WITH POWER “ON”. FOLLOW APPROPRIATE SAFETY PROCEDURES FOR WORKING WITH AN ENERGIZED SYSTEM.



NOTICE Follow ALL required safety and PPE procedures customary for your organization, when working on this product.

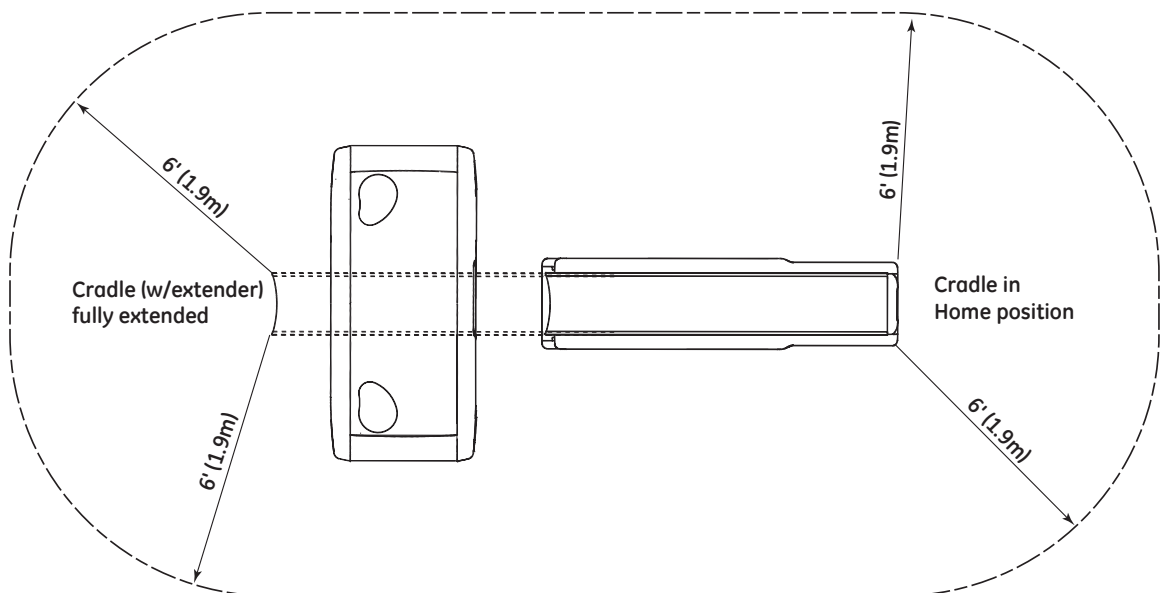
3.3 Required Conditions

- Only trained service personnel should service the GE CT Scanner.
- Footswitch cover must be removed.

Section 4.0 Procedure

4.1 Test Conditions

- Test with Table at maximum elevation and again with Table at minimum elevation
- Test to cover all points in an envelope described by Table travel from minimum to maximum extension (including table extender).
- Test from both the head and the foot of the Table -- e.g., test assuming patient feet first position as well as patient head first
- Test above (7ft or 2.1m) and below (6ft or 1.9m) Table surface -- e.g., test assuming patient lying face down as well as face up.
- Test for six feet (1.9m) envelope preventing patient access to all conductive surfaces including:
 - IV poles and tray assembly
 - any smart step monitors and stands
 - table bearing rails
 - all other conductive surfaces



4.2 Patient Touch Leakage Current Test Procedure



NOTICE These instructions have been validated only with a Dale 600/601 meter. Due to the unique nature of this meter, GE Healthcare cannot guarantee the accuracy of these procedures when performed with any meter other than the Dale 600/601.

- 1.) Move the table to ISO elevation.
- 2.) Remove the footswitch covers and the gantry left side cover.
- 3.) Refer to the Dale 601 Operator's Manual for instructions on the use of the Dale 600 meter for measuring leakage current (or refer to [Figure 4-1](#), for a quick overview).

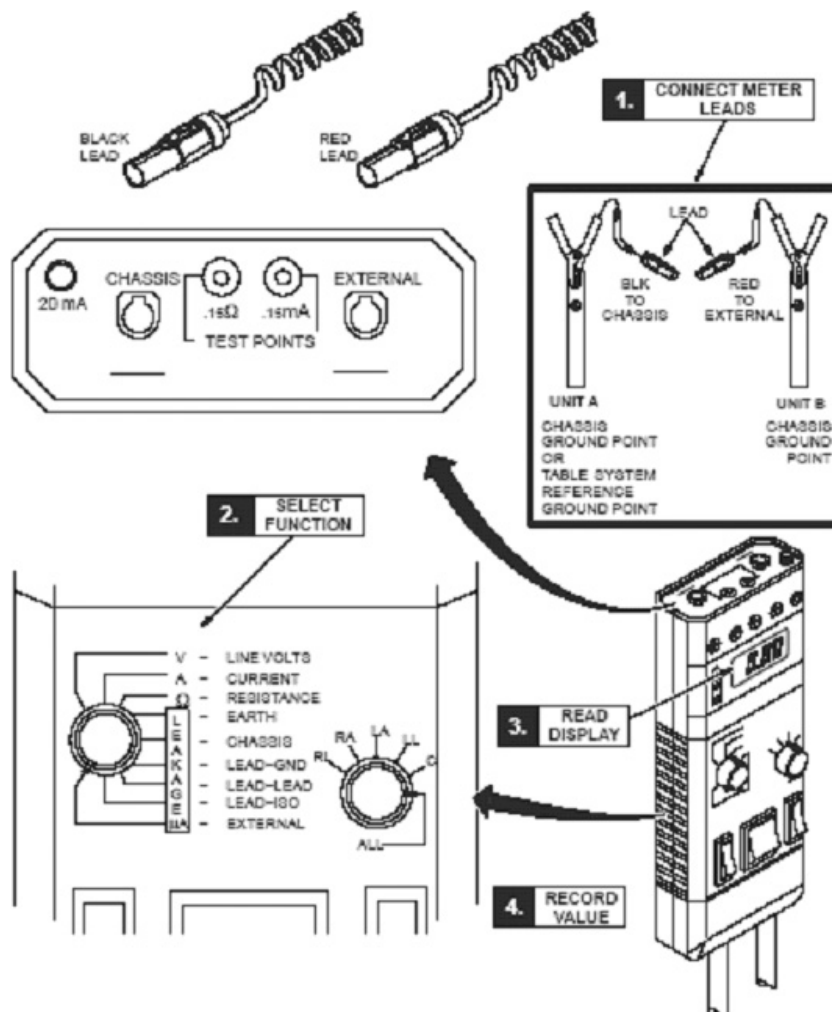


Figure 4-1 Using Dale 600 to Measure Leakage Current

- 4.) Plug the Dale 600 Meter into the outlet on the gantry left side. Confirm the outlet is correctly wired per the three LED indicators on the meter.
- 5.) Connect one end of the shorter black lead to the chassis plug and connect the other end to the table ground bus.
- 6.) Connect the longer red test lead (or the longer black lead) to the external plug on the top of the Dale 600 meter.
- 7.) Set the function switch on the Dale to external. Use the external lead to touch the meter's test terminal, to test that the meter is operational.

- 8.) The black lead will be connected to the table base ground bus, and the read lead will be connected to the devices (components) under test.
- Note:** Your meter may have two black leads that are keyed for chassis and the ground connection. A valid calibration sticker must be present on the meter used. Record this information on the GE 4879 form.
- 9.) Beware: Leakage current cannot exceed the following and is tested with power ON:
- Critical care areas (invasive) - 10µA CT Systems
 - General care areas - 20µA**
 - Not intended for patient area - 50µA
- 10.) Testing must be completed between the system reference ground point (table base) to unit reference ground points (i.e. gantry and table, see [Figure 4-3](#)).
- 11.) Test all conductive surfaces and components within patient reach or within 6 feet (1.9m) of the table and 7 feet (2.1m) above.
- Measure at table maximum travels.
 - At some sites, wall outlet cover plates and sinks may become an issue.
- 12.) Test all Optional components such as in-room monitors, injector, overhead monitor suspensions, and table options.
- Record these results in the data sheet. To be completed at install and during PM inspections.

Gnd Bus to	Install <10µA
Any table ground points within the 6' range	
Any gantry ground points within the 6' range	
Injector assembly metal surface	
Boom-in-Room metal surface	
Monitors or metal surface	
Sink or metal surface	
Installed table accessories	
(other)	
(other)	
(other)	



NOTICE

Be Aware of Static Discharge from: scan window, keypads, display, and touch pads or other plastic surfaces.

- 13.) **Procedure Hints:**
- Look for items that have abnormal measurements, high or low.
These could indicate mis-wiring, loose connections, or poor connections due to corrosion, painted surfaces, etc.
 - High leakage could indicate a wiring error such as a neutral connected to the ground.
 - Fluctuating ground currents could indicate a short, poor connection, or facilities ground problem causing leakage currents from other areas of the facility to flow through our system grounds.
 - Refer to the illustrations below, for measurement visual descriptions.

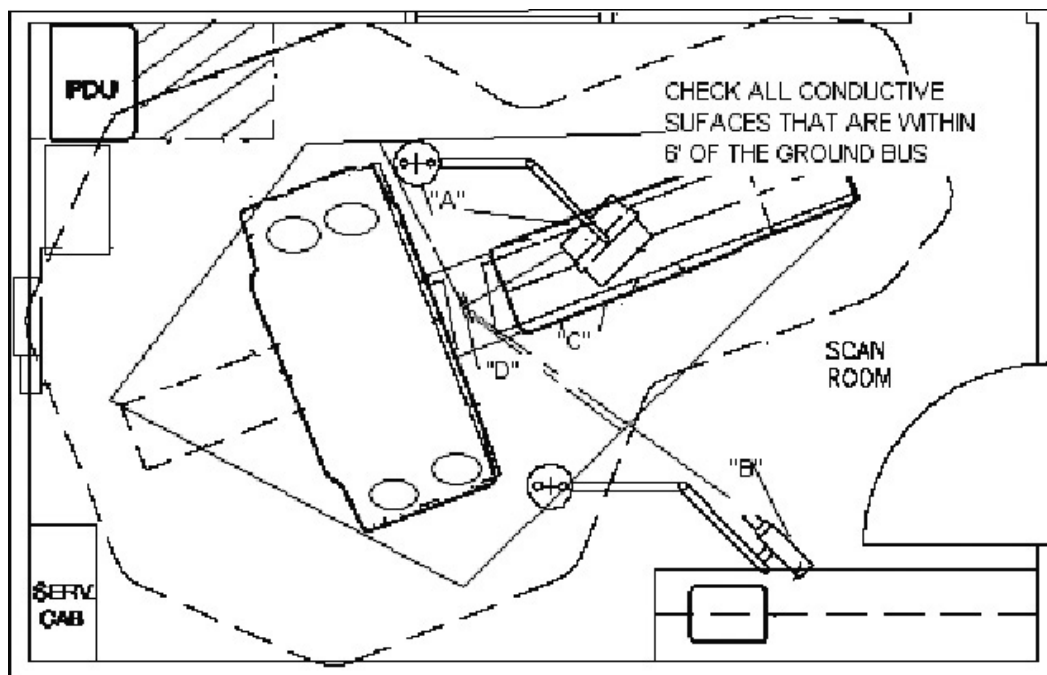


Figure 4-2 CT Leakage Current Map

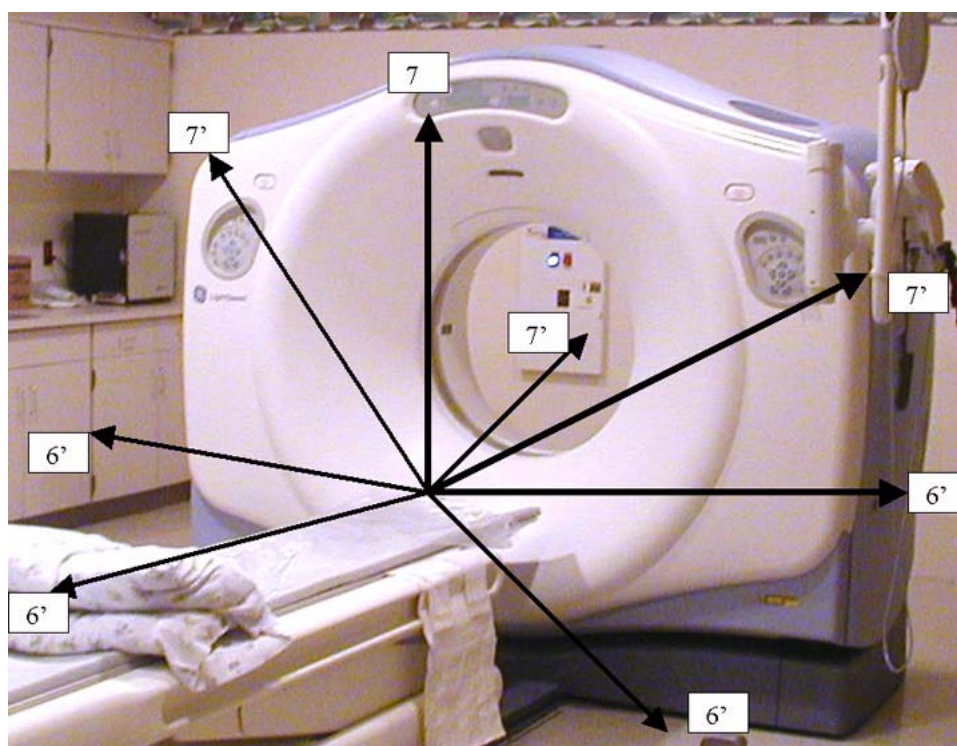


Figure 4-3 Measurement Points

4.3 Ground Resistance Checks Procedure (done during mechanical install)

Use Fluke 87 to measure ground resistance.

- 1.) Use a Fluke 87 or equivalent RMS Meter to verify the presence of less than 0.5 ohm of resistance between each of the following points:

FROM	TO	
PDU Ground Bus	Vault Ground	☐ Check box when complete
PDU Ground Bus	Table/Gantry raceway ground point	☐ Check box when complete
Table/Gantry raceway ground point	Gantry Chassis	☐ Check box when complete
Table/Gantry raceway ground point	Table Chassis	☐ Check box when complete
Table/Gantry raceway ground point	Operator Console Chassis	☐ Check box when complete
All Display or Computing Options (if any)	Operator Console Chassis	☐ Check box when complete

Table 4-1 Resistance Verification - Site Ground

- 2.) Connect the meter leads to the Fluke 87 meter.
- 3.) Set the function switch to resistance ohms.
- 4.) The black and red leads will be connected per Table 4-1.
- 5.) All ground resistance testing is done with power OFF. Ground current will cause errors in the resistance readings.
- 6.) Use lockout/tag out procedures.
- 7.) Maximum ground resistance is 0.5 ohms impedance.
- 8.) Using the table footswitch ground bar as a reference. Check the ground resistance to all 5 system grounds.

Record the measurements in the datasheet:

SYSTEM GROUND BUS	TYPICAL READING	SITE DATA
Gantry Ground Stud	0.5 Ohms	
Console Ground Stud	0.5 Ohms	
PDU Ground Stud	0.5 Ohms	
Table Ground Stud	0.5 Ohms	

Hints:

- Look for undersized ground wires and repair.
- Ground resistance readings should not fluctuate. If so this indicates an intermittent

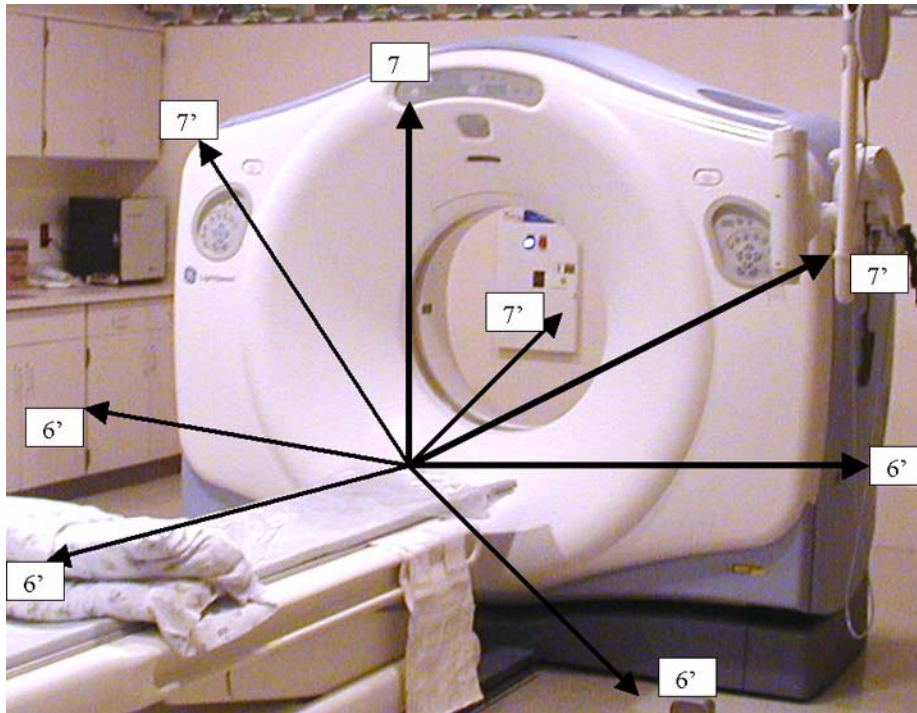


Figure 4-4 Measurement Points

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